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(54) **METHOD FOR TRANSMITTING AND RECEIVING SIGNALS IN WIRELESS COMMUNICATION SYSTEM, AND DEVICE SUPPORTING SAME**

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(57) **ABSTRACT**

An embodiment relates to a next-generation wireless communication system for supporting a higher data transmission rate, etc. than that of a 4th generation (4G) wireless communication system. According to an embodiment, a method for transmitting and receiving signals in a wireless communication system and a device supporting same may be provided, and another embodiment may be provided.

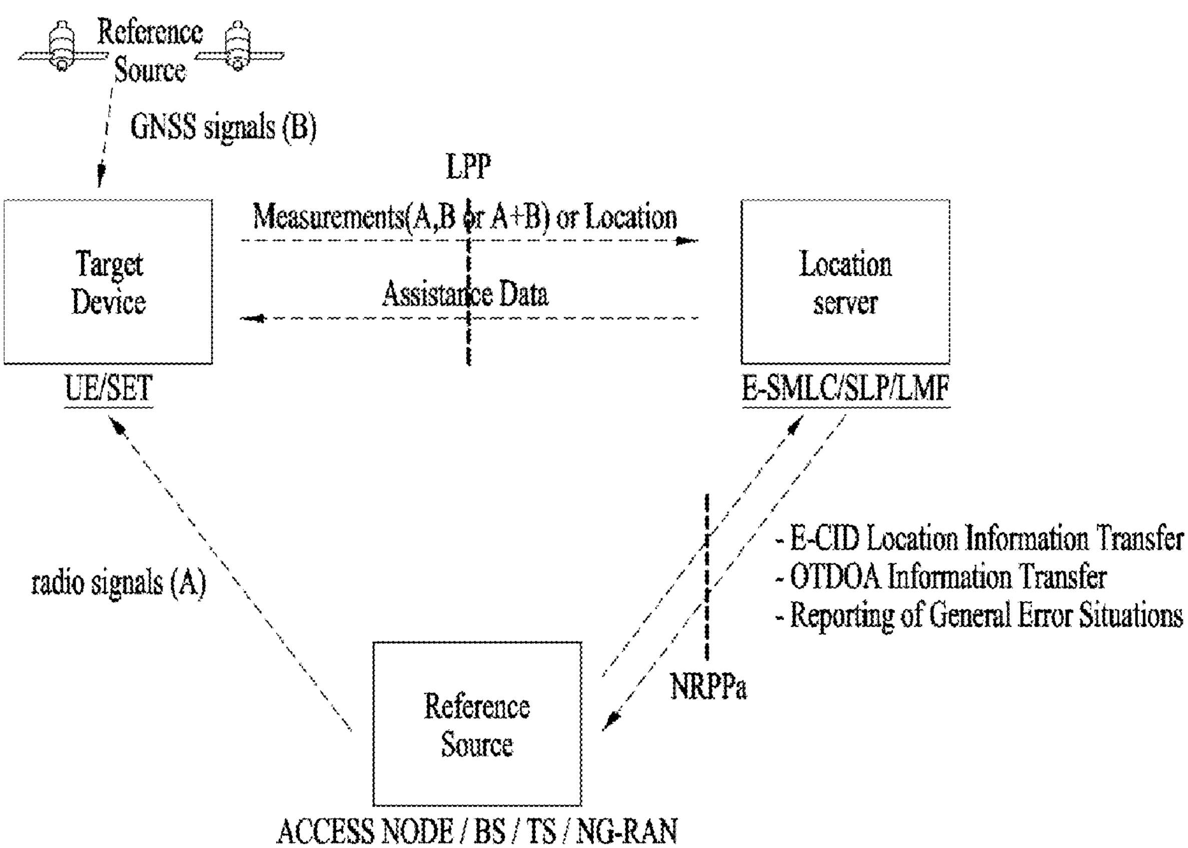


FIG. 1

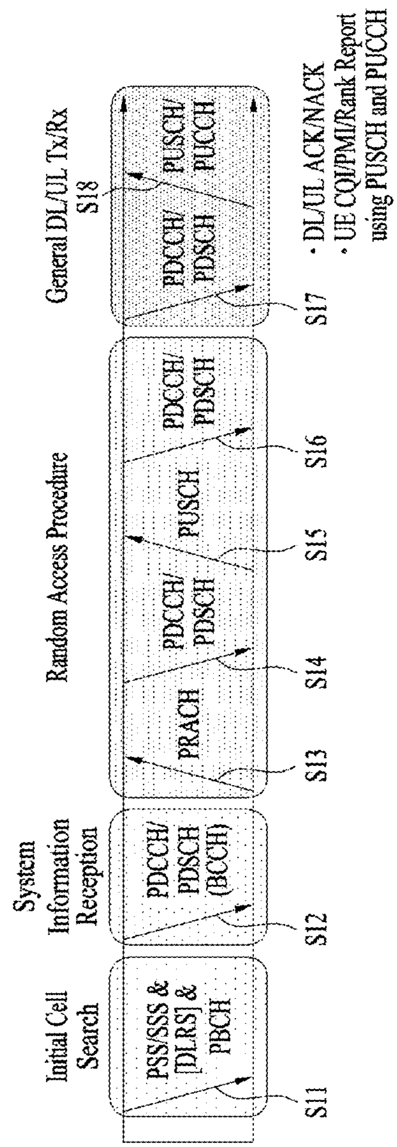
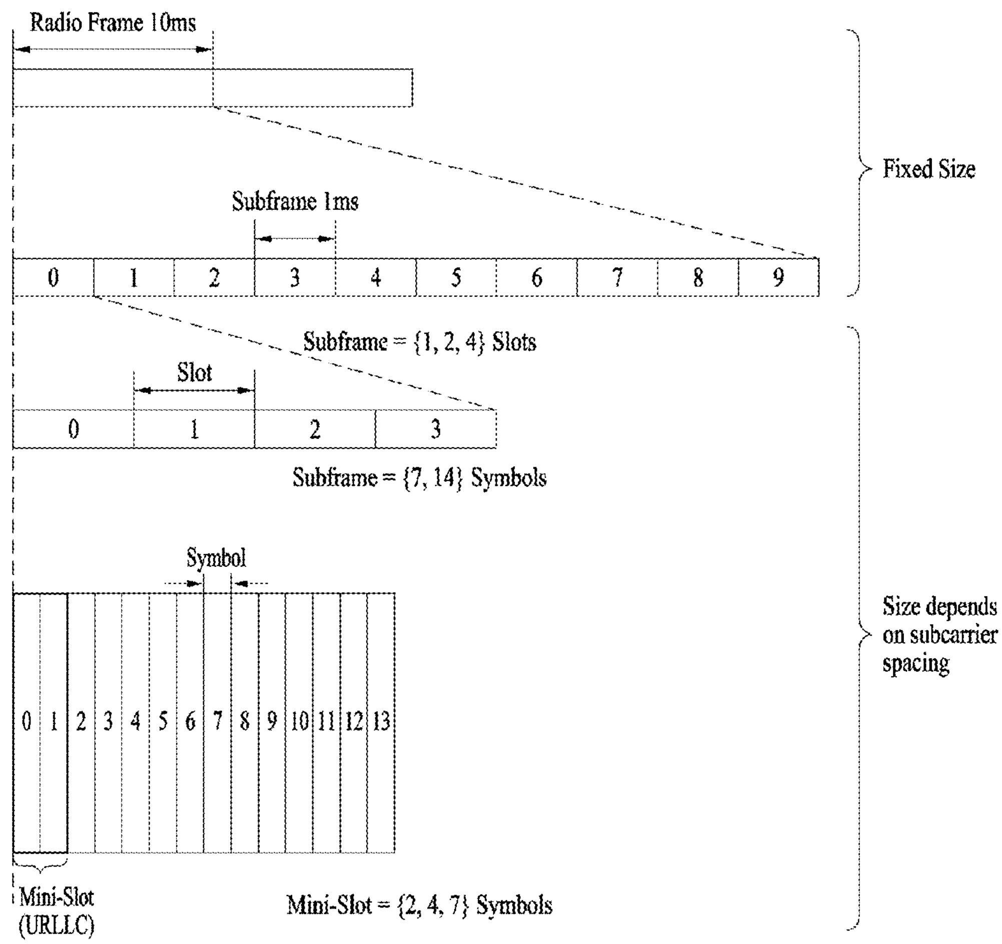

Initial Cell Search
PSS/SSS & [DL RS] & PBCH
S11
System Information Reception
PDCCH/ PDSCH (BCCH)
S12
Random Access Procedure
PRACH
S13
PDCCH/ PDSCH
S14
PUSCH
S15
PDCCH/ PDSCH
S16
General DL/UL Tx/Rx
PDCCH/ PDSCH
S17
S18
PUSCH/ PUCCH
· DL/UL ACK/NACK
· UE CQI/PMI/Rank Report using PUSCH and PUCCH

FIG. 2



Radio Frame 10ms
Fixed Size
Subframe 1ms
0
1
2
3
4
5
6
7
8
9
Subframe = {1, 2, 4} Slots
Slot
0
1
2
3
Subframe = {7, 14} Symbols
Symbol
0
1
2
3
4
5
6
7
8
9
10
11
12
13
Size depends on subcarrier spacing
Mini-Slot (URLLC)
Mini-Slot = {2, 4, 7} Symbols

FIG. 3

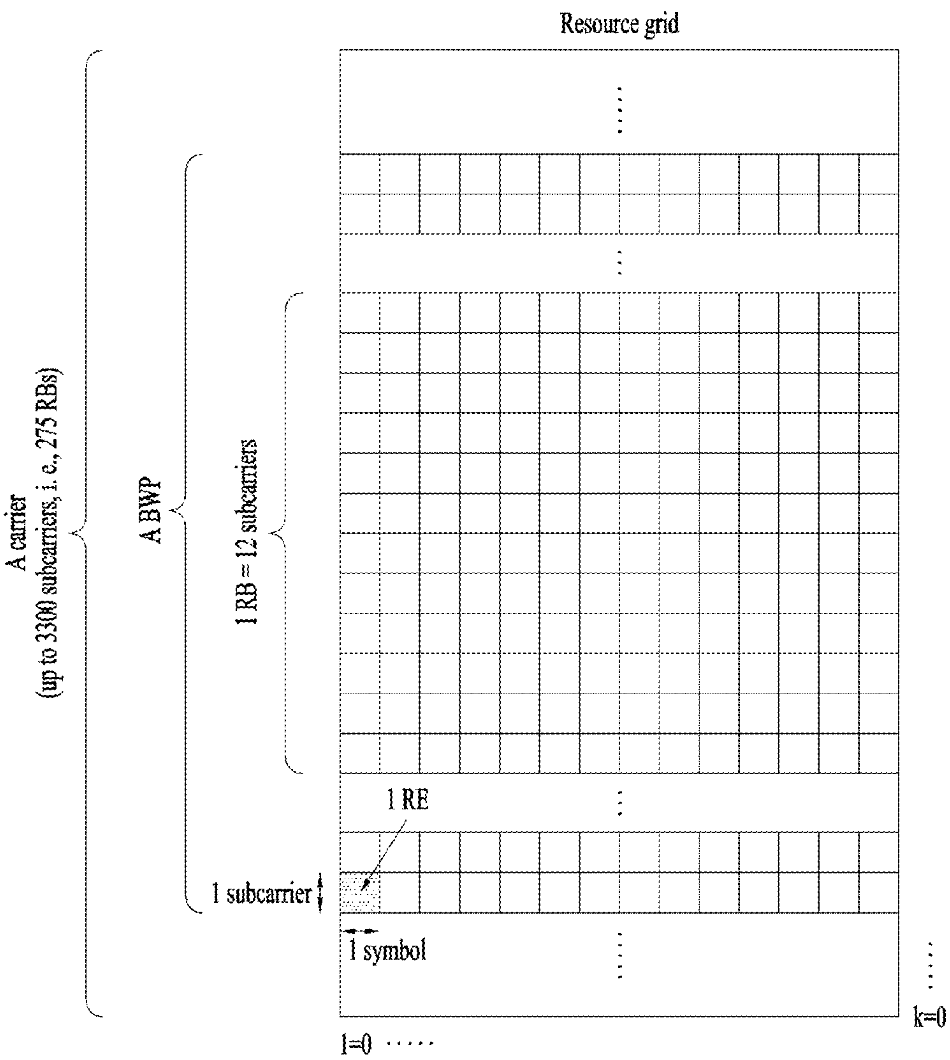

Resource grid
A carrier
(up to 3300 subcarriers, i. e., 275 RBs)
A BWP
1 RB = 12 subcarriers
1 RE
1 subcarrier
1 symbol
k=0
l=0

FIG. 4

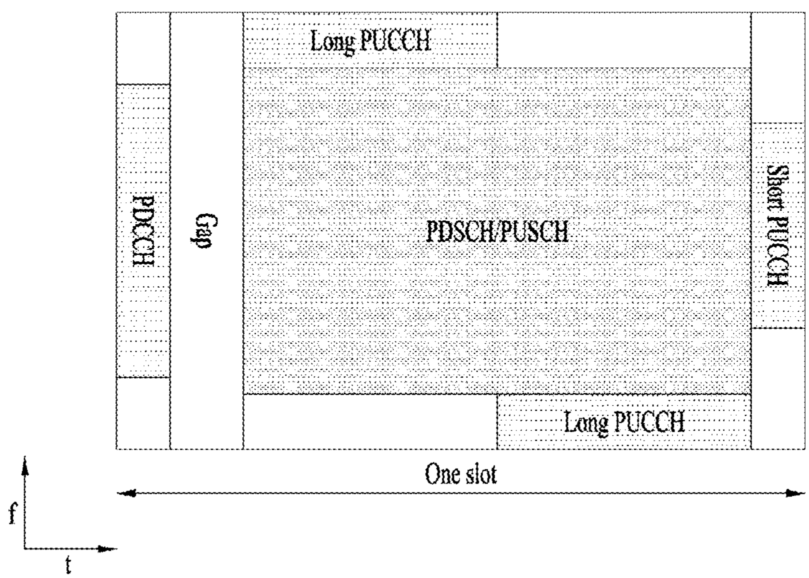
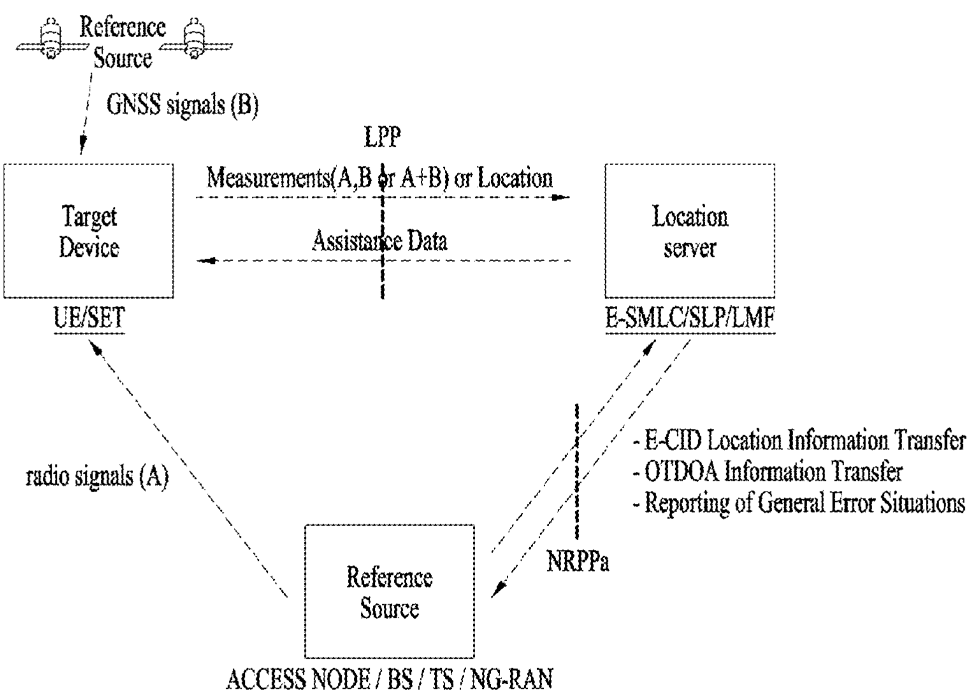

Long PUCCH
PDCCH
Gap
PDSCH/PUSCH
Short PUCCH
Long PUCCH
One slot
f
t

FIG. 5



Reference
Source
GNSS signals (B)
LPP
Measurements(A,B or A+B) or Location
Assistance Data
Target
Device
UE/SET
Location
server
E-SMLC/SLP/LMF
radio signals (A)
- E-CID Location Information Transfer
- OTDOA Information Transfer
- Reporting of General Error Situations
NRPPa
Reference
Source
ACCESS NODE / BS / TS / NG-RAN

FIG. 6

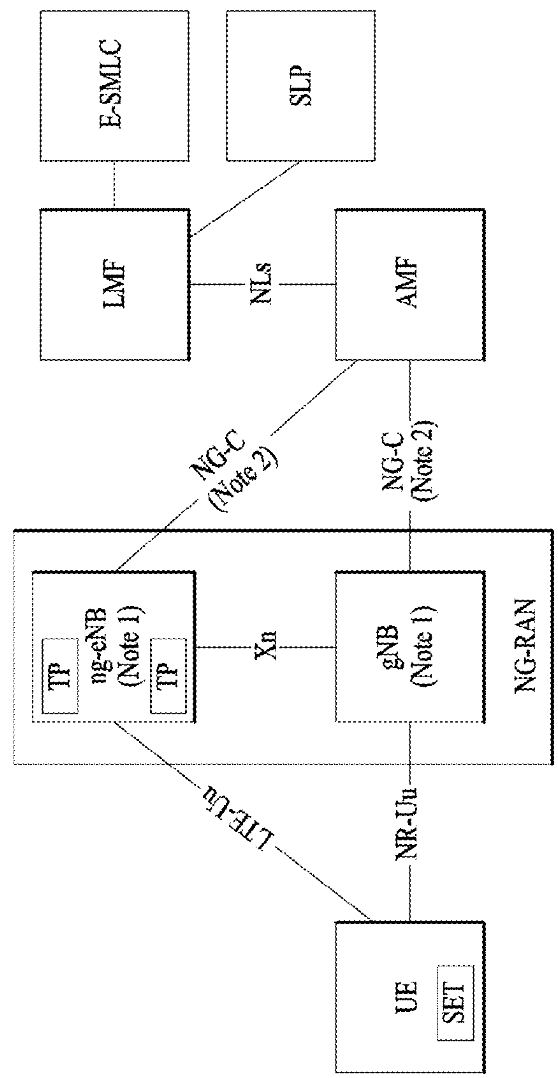

E-SMLC
SLP
LMF
NLs
AMF
NG-C (Note 2)
NG-C (Note 2)
TP
ng-eNB (Note 1)
TP
Xn
gNB (Note 1)
NG-RAN
LTE-Uu
NR-Uu
UE
SET

FIG. 7

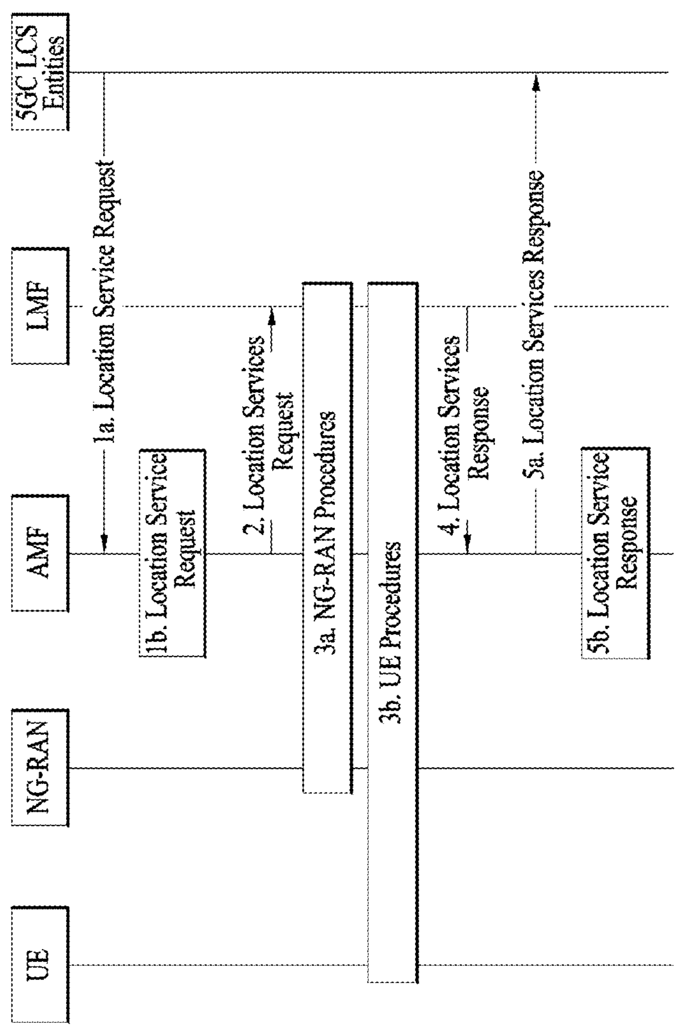

UE
NG-RAN
AMF
LMF
5GC LCS Entities
1a. Location Service Request
1b. Location Service Request
2. Location Services Request
3a. NG-RAN Procedures
3b. UE Procedures
4. Location Services Response
5a. Location Services Response
5b. Location Service Response

FIG. 8

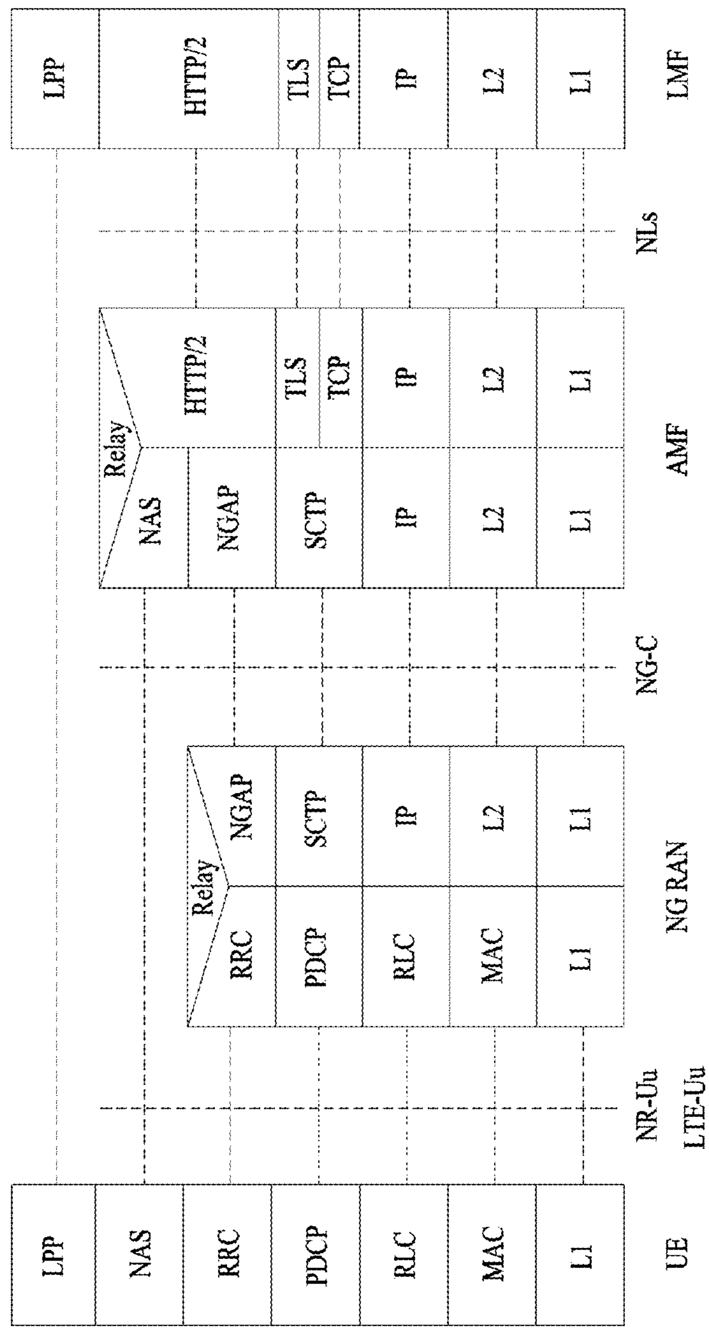

LPP
NAS
RRC
PDCP
RLC
MAC
L1
UE
NR-Uu
LTE-Uu
Relay
RRC
PDCP
RLC
MAC
L1
NGAP
SCTP
IP
L2
L1
NG RAN
NG-C
Relay
NAS
NGAP
SCTP
IP
L2
L1
HTTP/2
TLS
TCP
IP
L2
L1
AMF
NLs
LPP
HTTP/2
TLS
TCP
IP
L2
L1
LMF

FIG. 9

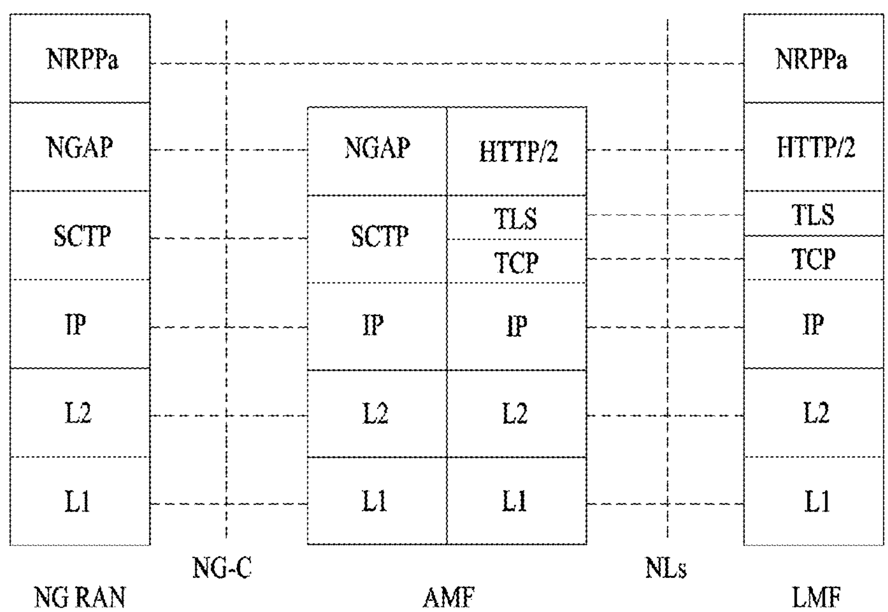

NRPPa
NGAP
SCTP
IP
L2
L1
NG RAN
NG-C
NGAP
HTTP/2
SCTP
TLS
TCP
IP
IP
L2
L2
L1
L1
AMF
NLs
NRPPa
HTTP/2
TLS
TCP
IP
L2
L1
LMF

FIG. 10

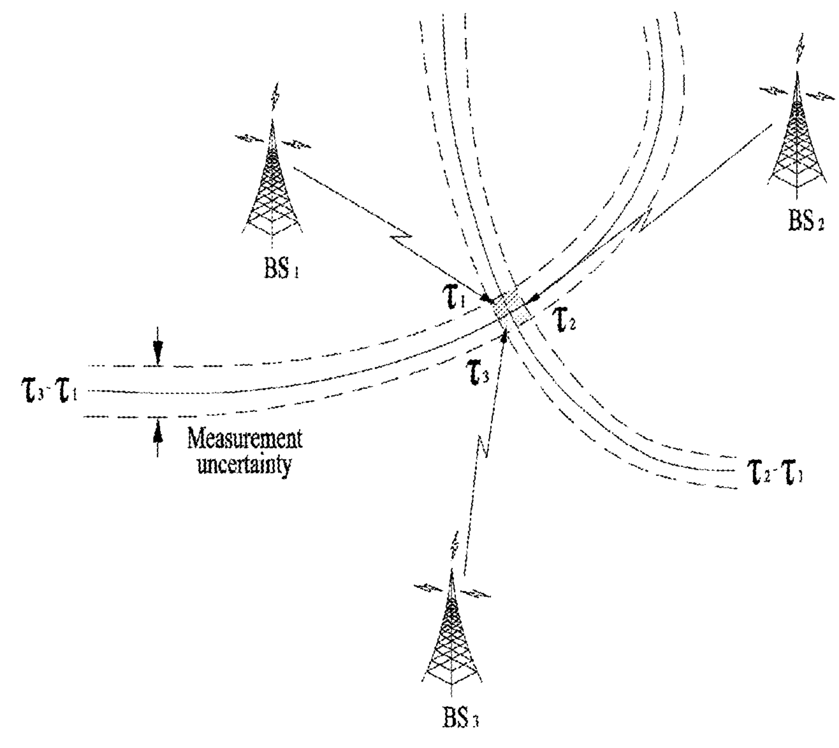
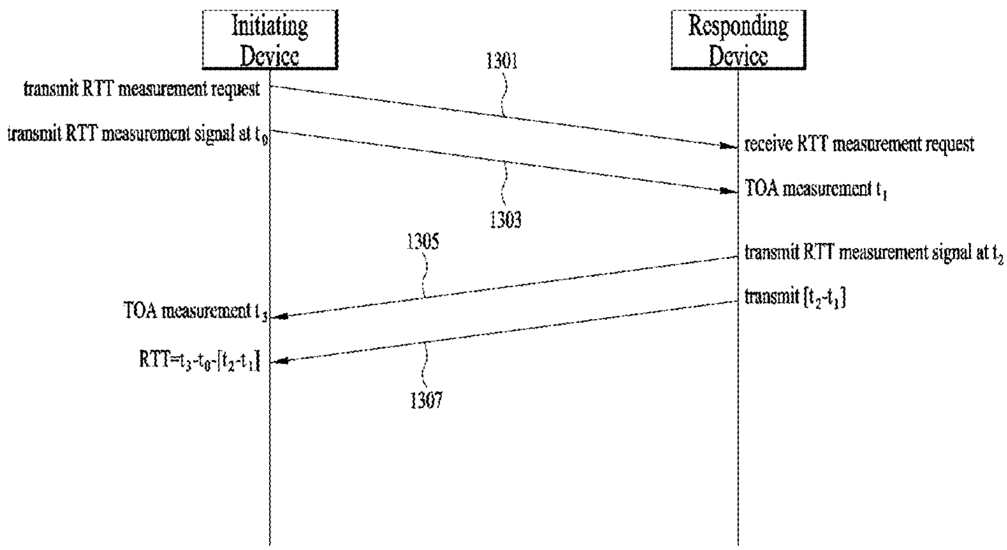
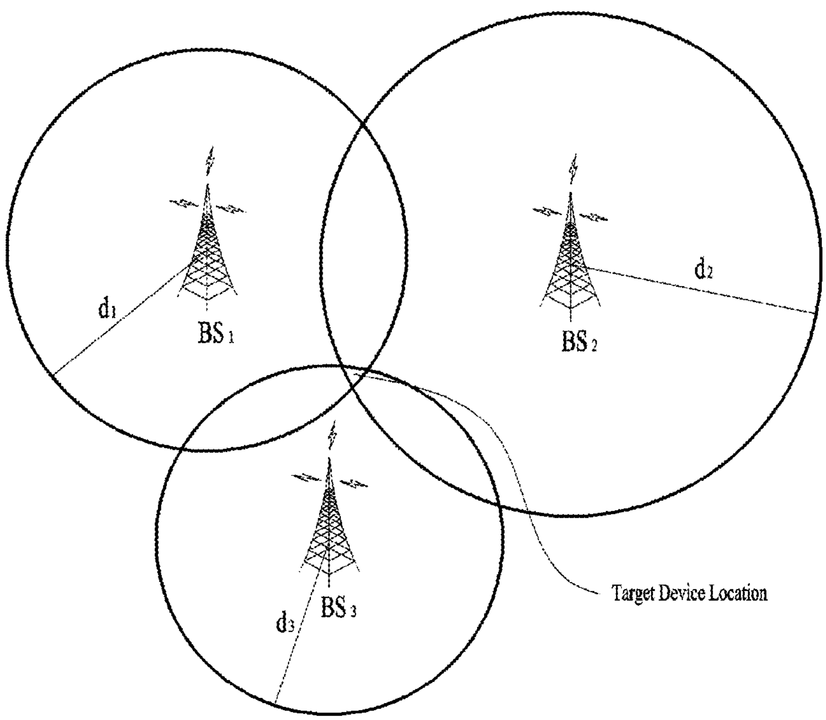


FIG. 11


Initiating Device
Responding Device
1301
transmit RTT measurement request
receive RTT measurement request
transmit RTT measurement signal at t_0
TOA measurement t_1
1303
1305
transmit RTT measurement signal at t_2
transmit $[t_2\text{-}t_1]$
TOA measurement t_3
$RTT=t_3\text{-}t_0\text{-}[t_2\text{-}t_1]$
1307

(a)


d_1
BS_1
d_2
BS_2
d_3
BS_3
Target Device Location

(b)

FIG. 12

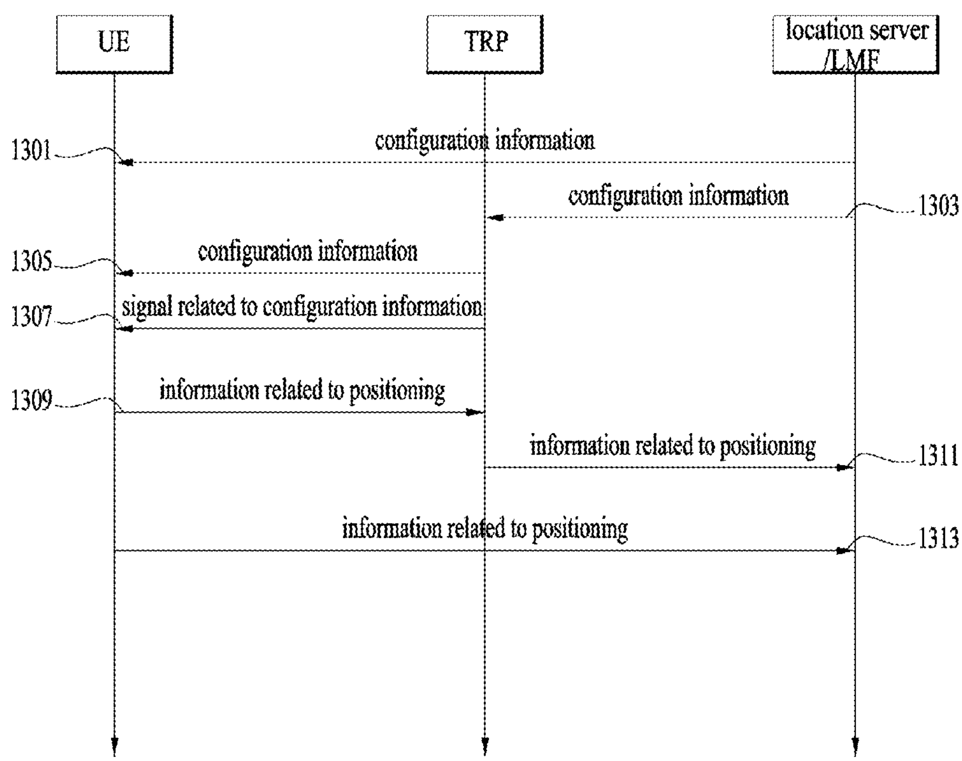

UE
TRP
location server /LMF
1301
configuration information
configuration information
1303
1305
configuration information
1307
signal related to configuration information
1309
information related to positioning
information related to positioning
1311
information related to positioning
1313

FIG. 13

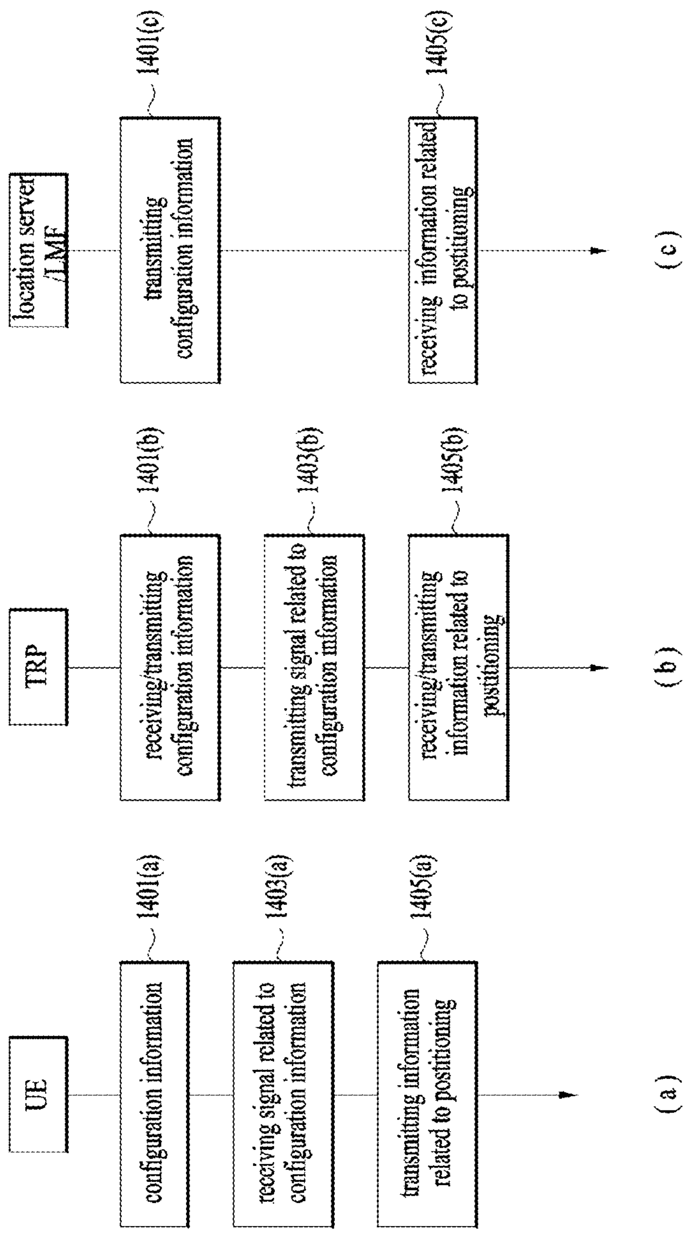

UE
configuration information
1401(a)
receiving signal related to configuration information
1403(a)
transmitting information related to positioning
1405(a)
(a)
TRP
receiving/transmitting configuration information
1401(b)
transmitting signal related to configuration information
1403(b)
receiving/transmitting information related to positioning
1405(b)
(b)
location server /LMF
transmitting configuration information
1401(c)
receiving information related to positioning
1405(c)
(c)

FIG. 14

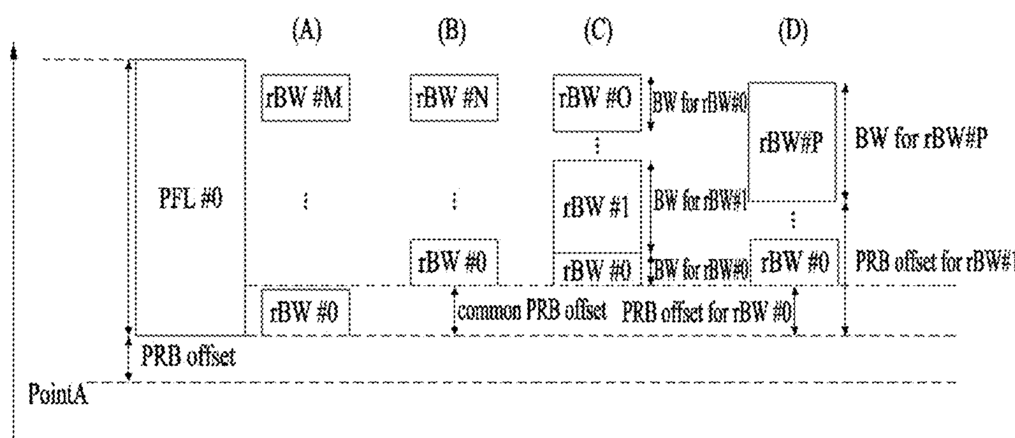

(A)
(B)
(C)
(D)
rBW #M
rBW #N
rBW #O
BW for rBW#0
rBW#P
BW for rBW#P
PFL #0
rBW #1
BW for rBW#1
rBW #0
rBW #0
BW for rBW#0
rBW #0
PRB offset for rBW#1
rBW #0
common PRB offset
PRB offset for rBW #0
PRB offset
PointA

FIG. 15

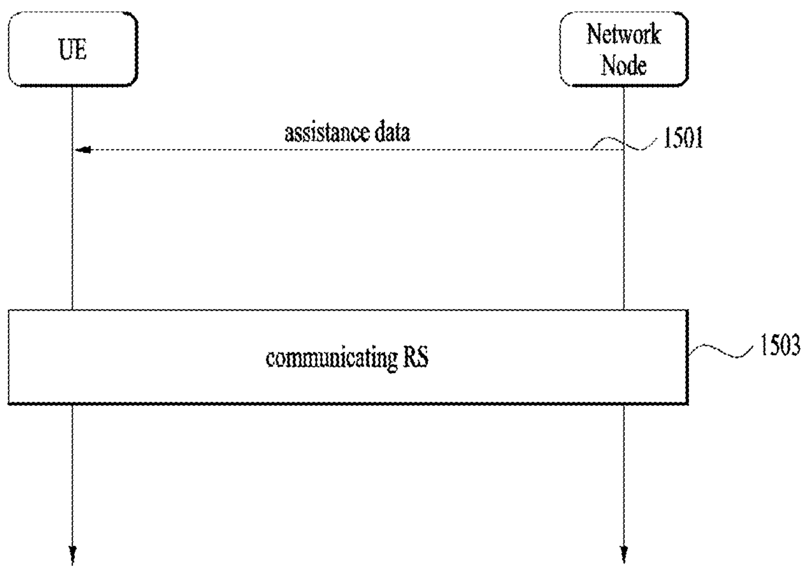

UE
Network Node
assistance data
1501
communicating RS
1503

FIG. 16

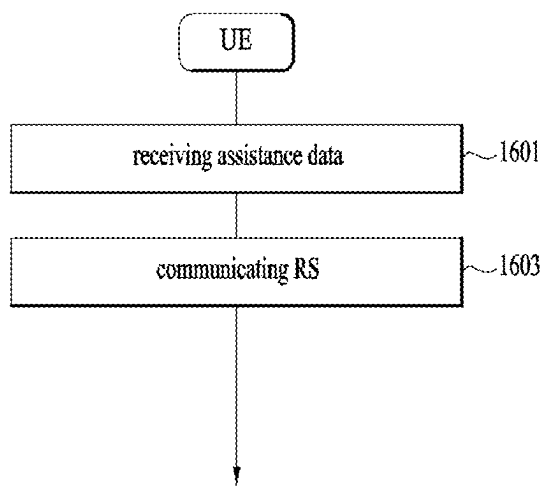


FIG. 17

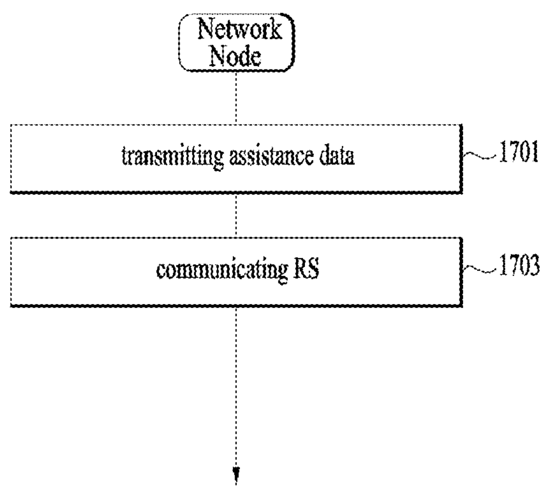

Network Node
transmitting assistance data
1701
communicating RS
1703

FIG. 18

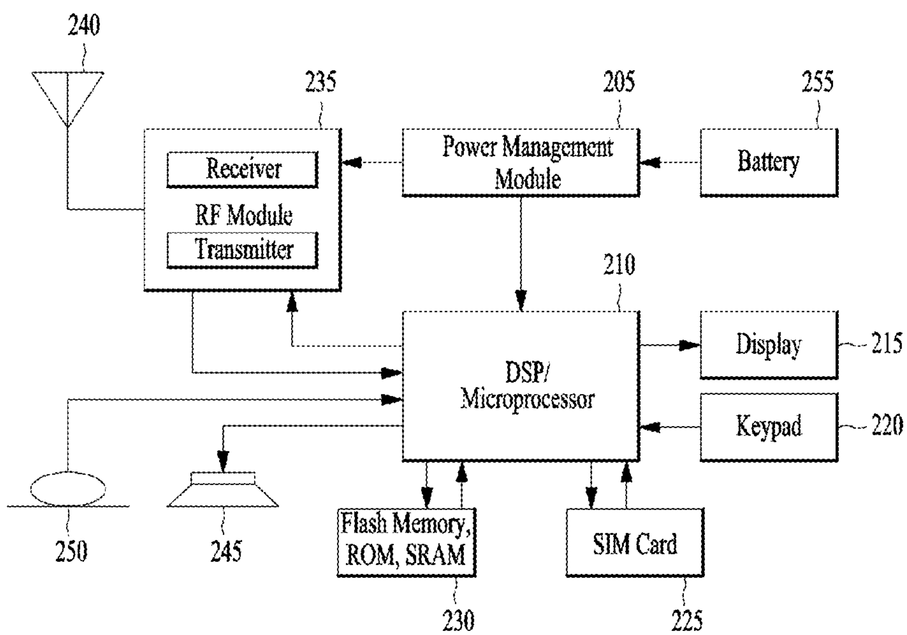

240
235
205
255
Receiver
RF Module
Transmitter
Power Management Module
Battery
210
DSP/ Microprocessor
Display
215
Keypad
220
Flash Memory, ROM, SRAM
SIM Card
250
245
230
225

FIG. 19

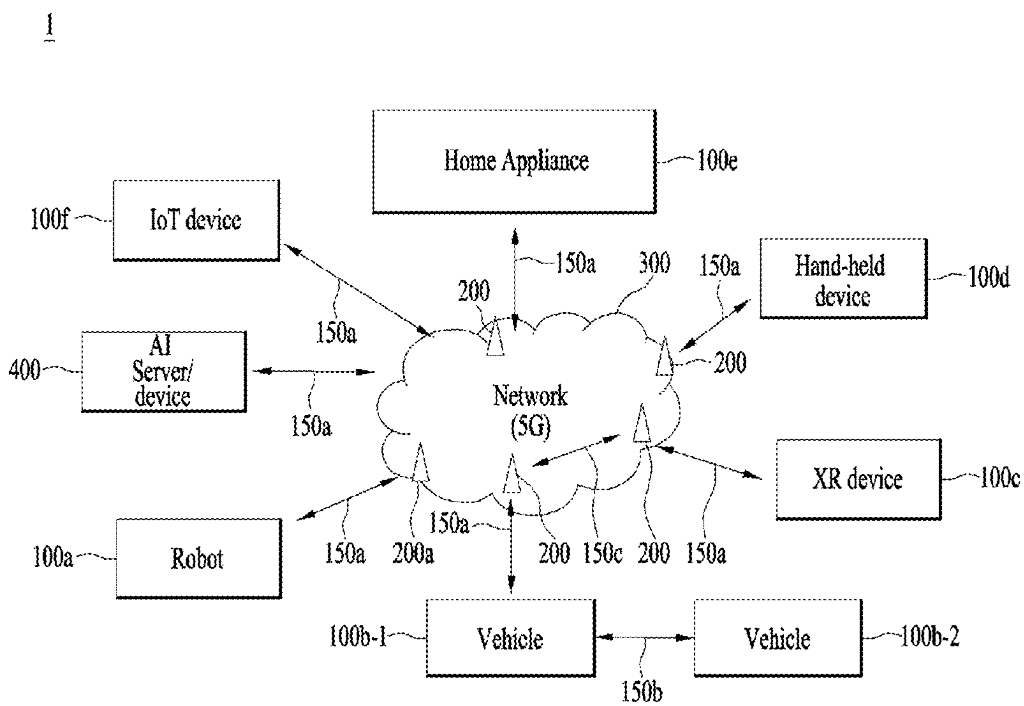

1
Home Appliance
100e
100f
IoT device
150a
300
150a
Hand-held device
100d
200
150a
400
AI Server/ device
200
Network (5G)
150a
XR device
100c
100a
Robot
150a
200a
150a
200
150c
200
150a
100b-1
Vehicle
Vehicle
100b-2
150b

FIG. 20

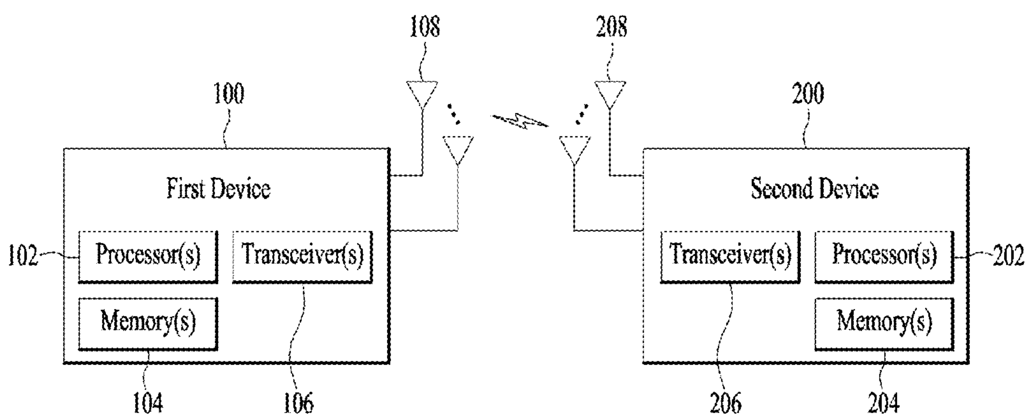

108
208
100
200
First Device
Second Device
102
Processor(s)
Transceiver(s)
Transceiver(s)
Processor(s)
202
Memory(s)
Memory(s)
104
106
206
204

FIG. 21

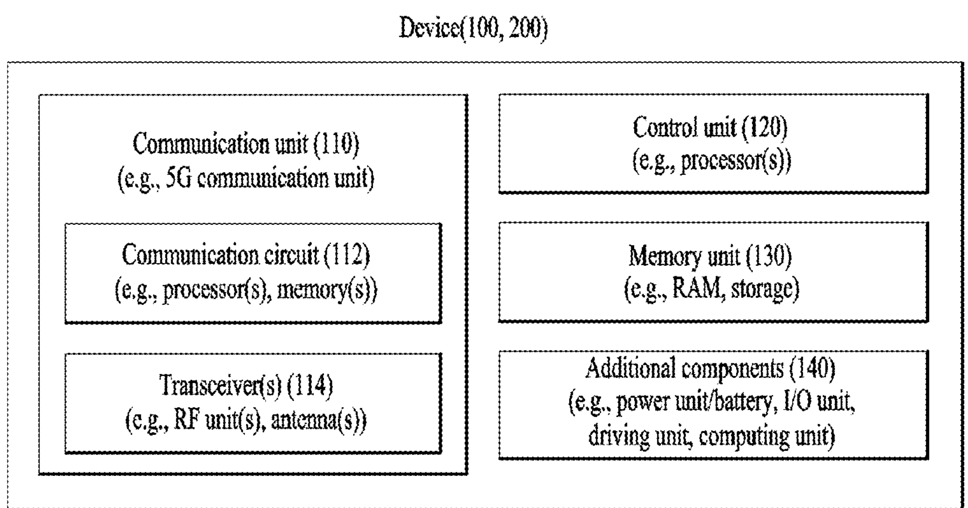

Device(100, 200)
Communication unit (110)
(e.g., 5G communication unit)
Communication circuit (112)
(e.g., processor(s), memory(s))
Transceiver(s) (114)
(e.g., RF unit(s), antenna(s))
Control unit (120)
(e.g., processor(s))
Memory unit (130)
(e.g., RAM, storage)
Additional components (140)
(e.g., power unit/battery, I/O unit,
driving unit, computing unit)

FIG. 22

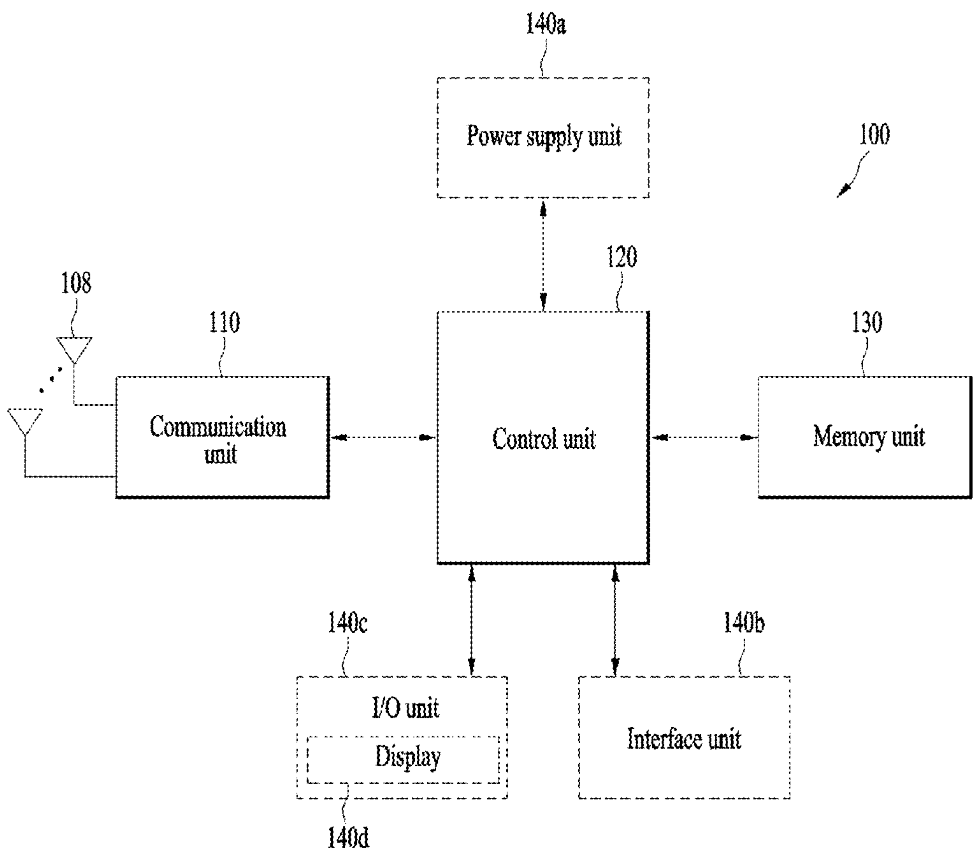
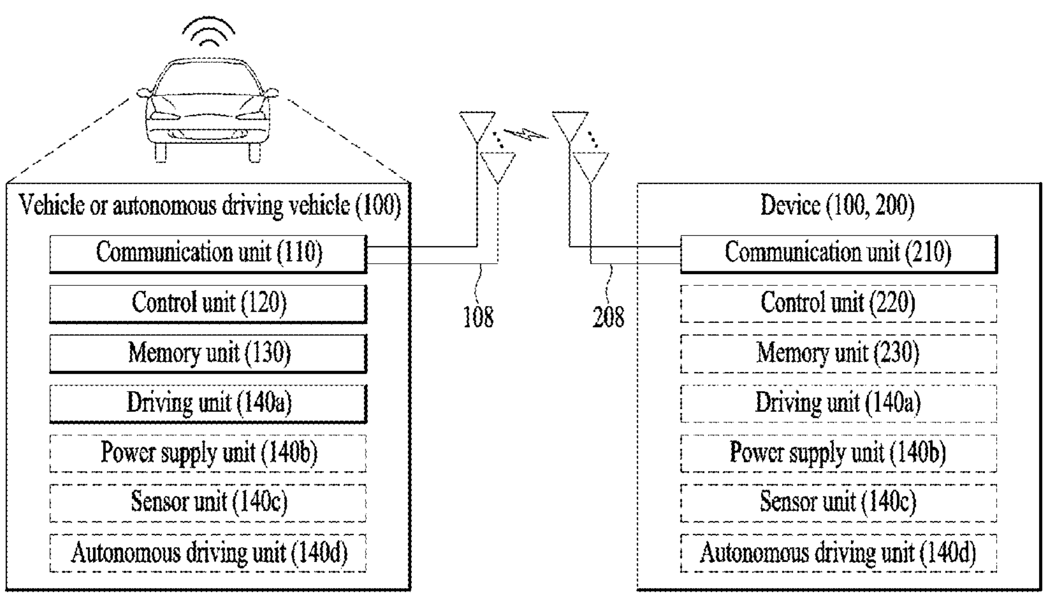

140a
Power supply unit
100
120
108
110
130
Communication unit
Control unit
Memory unit
140c
140b
I/O unit
Display
Interface unit
140d

FIG. 23


Vehicle or autonomous driving vehicle (100)
Communication unit (110)
Control unit (120)
Memory unit (130)
Driving unit (140a)
Power supply unit (140b)
Sensor unit (140c)
Autonomous driving unit (140d)
108
208
Device (100, 200)
Communication unit (210)
Control unit (220)
Memory unit (230)
Driving unit (140a)
Power supply unit (140b)
Sensor unit (140c)
Autonomous driving unit (140d)

METHOD FOR TRANSMITTING AND RECEIVING SIGNALS IN WIRELESS COMMUNICATION SYSTEM, AND DEVICE SUPPORTING SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2022/018809, filed on Nov. 25, 2022, which claims the benefit of earlier filing date and right of priority to Korean Application No. 10-2022-0053213, filed on Apr. 29, 2022, the contents of which are all hereby incorporated by reference herein in their entireties.

TECHNICAL FIELD

[0002] One embodiment relates to a wireless communication system.

BACKGROUND ART

[0003] As a number of communication devices have required higher communication capacity, the necessity of the mobile broadband communication much improved than the existing radio access technology (RAT) has increased. In addition, massive machine type communications (MTC) capable of providing various services at anytime and anywhere by connecting a number of devices or things to each other has been considered in the next generation communication system. Moreover, a communication system design capable of supporting services/UEs sensitive to reliability and latency has been discussed.

SUMMARY

[0004] An object of the present disclosure is to provide a method for transmitting and receiving signals in a wireless communication system and a device supporting the same.

[0005] To achieve these objects and other advantages and in accordance with the purpose of the disclosure, as embodied and broadly described herein, there is provided a method and apparatus for transmitting and receiving a signal in a wireless communication system.

[0006] In accordance with one embodiment, a method for transmitting and receiving signals in a wireless communication system and an apparatus for supporting the same are provided.

[0007] In accordance with one embodiment, a method supported by a user equipment (UE) in a wireless communication system is provided.

[0008] The method performed by the user equipment (UE) in a wireless communication system may include: receiving assistance data including information related to a configuration for a reference signal (RS) for positioning.

[0009] The information related to the configuration for the RS may include information related to one or more positioning frequency layers (PFLs).

[0010] The method may further include performing communication for the RS for positioning based on the assistance data.

[0011] Based on information indicating that the UE is a reduced capability (RedCap) UE, i) the information related to the RS configuration further includes information related to a plurality of first bandwidths (BWs) for the RS; and ii) the RS is communicated in a single first specific bandwidth (BW) among the plurality of first BWs at a single first time instance.

[0012] The plurality of first BWs may be included in one first PFL among the one or more PFLs in a frequency domain.

[0013] Based on information indicating that the UE is the RedCap UE and BW-based frequency hopping is configured, the RS is communicated in the first specific BW at the first time instance, and is communicated in one second specific BW different from the first specific BW among the plurality of first BWs at one second time instance based on the BW-based frequency hopping.

[0014] The second time instance is configured at a time later than the first time instance in a time domain.

[0015] The plurality of first bandwidths (BWs) may be related to a plurality of BW identifiers (IDs).

[0016] In the BW-based frequency hopping, the first specific BW and the second specific BW are identified according to first pattern information based on the plurality of BW IDs.

[0017] Based on the information indicating that the UE is the RedCap UE, the information related to the RS configuration may further include information related to a plurality of second BWs for the RS.

[0018] The plurality of second BWs may be included in the second PFL different from the first PFL among the one or more PFLs within the frequency domain.

[0019] Based on information indicating that the UE is the RedCap UE and BW-based frequency hopping is configured, i) the RS is communicated in the first specific BW at the first time instance, and ii) the RS is communicated in one third specific BW among the plurality of first BWs at one third time instance based on the PFL-based frequency hopping.

[0020] The third time instance may be configured at a time later than the first time instance in a time domain.

[0021] The one or more PFLs may be associated with one or more PFL IDs.

[0022] In the PFL-based frequency hopping, the first PFL and the second PFL may be identified according to second pattern information based on the one or more PFL IDs.

[0023] A capability report related to capability of the UE is transmitted.

[0024] Based on the information indicating that the UE is the RedCap UE, the capability report may include information related to a maximum number of a plurality of BWs for the RS supportable by the RedCap UE.

[0025] The number of the plurality of first BWs and the number of the plurality of second BWs are set to be less than or equal to the maximum number.

[0026] The plurality of first BWs may be configured based on at least one of i), ii), iii) and iv); wherein i) means that the plurality of first BWs, in the frequency domain, starts from a lowest physical resource block (PRB) of the first PFL and has a same size; ii) means that the plurality of first BWs, in the frequency domain, starts from a starting point that is identified based on a common PRB offset configured based on the lowest PRB of the first PFL; iii) means that the plurality of first BWs, in the frequency domain, starts a starting point that is identified based on the common PRB offset configured based on the lowest PRB of the first PFL, and is configured based on information about a plurality of BW sizes for the plurality of first BWs; and iv) means that the plurality of first BWs, in the frequency domain, is configured not only based on a plurality of PRB offsets for the plurality of first BWs configured based on the lowest PRB of the first PFL, but also based on information about the plurality of BW sizes for the plurality of first BWs, and is also configured to have the same size.

[0027] The RS may include at least one of a positioning reference signal (PRS) and a sounding reference signal (SRS) for the positioning.

[0028] In accordance with one embodiment, a user equipment (UE) operated in a wireless communication system is provided.

[0029] In accordance with one embodiment, a user equipment (UE) operated in a wireless communication system may include: a transceiver; and at least one processor connected to the transceiver.

[0030] The at least one processor may receive assistance data including information related to a configuration for a reference signal (RS) for positioning.

[0031] The information related to the configuration for the RS includes information related to one or more positioning frequency layers (PFLs).

[0032] The at least one processor may perform communication for the RS for positioning based on the assistance data.

[0033] Based on information indicating that the UE is a reduced capability (RedCap) UE, i) the information related to the RS configuration further includes information related to a plurality of bandwidths (BWs) for the RS; and ii) the RS is communicated in a single first specific bandwidth (BW) among the plurality of first BWs at a single first time instance.

[0034] The plurality of first BWs may be included in one first PFL among the one or more PFLs in a frequency domain.

[0035] The plurality of first BWs may be configured based on at least one of i), ii), iii) and iv); wherein i) means that the plurality of first BWs, in the frequency domain, starts from a lowest physical resource block (PRB) of the first PFL and has a same size; ii) means that the plurality of first BWs, in the frequency domain, starts from a starting point that is identified based on a common PRB offset configured based on the lowest PRB of the first PFL; iii) means that the plurality of first BWs, in the frequency domain, starts a starting point that is identified based on the common PRB offset configured based on the lowest PRB of the first PFL, and is configured based on information about a plurality of BW sizes for the plurality of first BWs; and iv) means that the plurality of first BWs, in the frequency domain, is configured not only based on a plurality of PRB offsets for the plurality of first BWs configured based on the lowest PRB of the first PFL, but also based on information about the plurality of BW sizes for the plurality of first BWs, and is also configured to have the same size.

[0036] The at least one processor is configured to communicate with at least one of a mobile terminal, a network, and an autonomous vehicle other than a vehicle including the UE.

[0037] In accordance with another aspect of the present disclosure, a method performed by a base station in a wireless communication system is provided.

[0038] A method performed by a base station (BS) in a wireless communication system may include: transmitting assistance data including information related to a configuration for a reference signal (RS) for positioning.

[0039] The information related to the configuration for the RS includes information related to one or more positioning frequency layers (PFLs).

[0040] The method may include performing communication for the RS for positioning related to the assistance data with a user equipment (UE).

[0041] Based on information indicating that the UE is a reduced capability (RedCap) UE, i) the information related to the RS configuration further includes information related to a plurality of bandwidths (BWs) for the RS; and ii) the RS is communicated in a single first specific bandwidth (BW) among the plurality of first BWs at a single first time instance.

[0042] The plurality of first BWs is included in one first PFL among the one or more PFLs in a frequency domain.

[0043] In accordance with another aspect of the present disclosure, a base station operated in a wireless communication system is provided.

[0044] The base station may include a transceiver; and at least one processor connected to the transceiver.

[0045] The at least one processor is configured to: transmit assistance data including information related to a configuration for a reference signal (RS) for positioning.

[0046] The information related to the configuration for the RS includes information related to one or more positioning frequency layers (PFLs).

[0047] The at least one processor may perform communication for the RS for positioning related to the assistance data with a user equipment (UE).

[0048] Based on information indicating that the UE is a reduced capability (RedCap) UE, i) the information related to the RS configuration further includes information related to a plurality of bandwidths (BWs) for the RS; and ii) the RS is communicated in a single first specific bandwidth (BW) among the plurality of first BWs at a single first time instance.

[0049] The plurality of first BWs may be included in one first PFL among the one or more PFLs in a frequency domain.

[0050] In accordance with another aspect of the present disclosure, an apparatus operated in a wireless communication system is provided.

[0051] The apparatus may include at least one processor; and at least one memory operatively connected to the at least one processor and configured to store one or more instructions such that the at least one processor performs specific operations by executing the instructions.

[0052] The specific operations may include: receiving assistance data including information related to a configuration for a reference signal (RS) for positioning.

[0053] The information related to the configuration for the RS includes information related to one or more positioning frequency layers (PFLs).

[0054] The specific operations may include performing communication for the RS for positioning based on the assistance data,

[0055] Based on information indicating that the UE is a reduced capability (RedCap) UE, i) the information related to the RS configuration further includes information related to a plurality of bandwidths (BWs) for the RS; and ii) the RS is communicated in a single first specific bandwidth (BW) among the plurality of first BWs at a single first time instance.

[0056] The plurality of first BWs may be included in one first PFL among the one or more PFLs in a frequency domain.

[0057] In accordance with another aspect of the present disclosure, a non-transitory processor-readable medium configured to store at least one instruction that allows at least one processor to perform specific operations by executing the instructions may include performing the specific operations.

[0058] The specific operations include receiving assistance data including information related to a configuration for a reference signal (RS) for positioning.

[0059] The information related to the configuration for the RS includes information related to one or more positioning frequency layers (PFLs).

[0060] The specific operations include performing communication for the RS for positioning based on the assistance data.

[0061] Based on information indicating that the UE is a reduced capability (RedCap) UE, i) the information related to the RS configuration further includes information related to a plurality of bandwidths (BWs) for the RS; and ii) the RS is communicated in a single first specific bandwidth (BW) among the plurality of first BWs at a single first time instance.

[0062] The plurality of first BWs may be included in one first PFL among the one or more PFLs in a frequency domain.

[0063] The above-described aspects of the present disclosure are only some of the preferred embodiments of the present disclosure, and various embodiments reflecting the technical features of the present disclosure may be derived and understood from the following detailed description of the present disclosure by those skilled in the art.

[0064] As is apparent from the above description, signals can be effectively transmitted and received in a wireless communication system.

[0065] According to one embodiment, positioning can be effectively performed in a wireless communication system.

[0066] According to one embodiment, the accuracy of positioning with respect to a reduced capability (RedCap) UE can be improved.

[0067] According to one embodiment, a base station (BS)/ location management function (LMF) can simultaneously support legacy UEs and a RedCap UEs in relation to such positioning.

[0068] It is to be understood that both the foregoing general description and the following detailed description of the present disclosure are exemplary and explanatory and are intended to provide further explanation of the disclosure as claimed.

DESCRIPTION OF DRAWINGS

[0069] The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this application, illustrate embodiment(s) of the disclosure and together with the description serve to explain the principle of the disclosure.

[0070] FIG. **1** is a diagram illustrating physical channels and a signal transmission method using the physical channels, which may be used in the embodiment.

[0071] FIG. **2** is a diagram illustrating a radio frame structure in a new radio access technology (NR) system to which the embodiment is applicable.

[0072] FIG. **3** is a diagram illustrating mapping of physical channels in a slot, to which the embodiment is applicable.

[0073] FIG. **4** is a diagram illustrating mapping of physical channels in a slot, to which the embodiment is applicable.

[0074] FIG. **5** is a diagram illustrating an exemplary positioning protocol configuration for user equipment (UE) positioning, which is applicable to the embodiment.

[0075] FIG. **6** is a diagram illustrating an example of an architecture of a system for positioning a UE, to which the embodiment is applicable.

[0076] FIG. **7** is a diagram illustrating an example of a procedure of positioning a UE, to which the embodiment is applicable.

[0077] FIG. **8** is a diagram illustrating protocol layers for supporting LTE positioning protocol (LPP) message transmission, to which the embodiment is applicable.

[0078] FIG. **9** is a diagram illustrating protocol layers for supporting NR positioning protocol A (NRPPa) protocol data unit (PDU) transmission, to which the embodiment is applicable.

[0079] FIG. **10** is a diagram illustrating an observed time difference of arrival (OTDOA) positioning method, to which the embodiment is applicable.

[0080] FIG. **11** is a diagram illustrating a multi-round trip time (multi-RTT) positioning method to which the embodiment is applicable.

[0081] FIG. **12** is a simplified diagram illustrating a method of operating a UE, a transmission and reception point (TRP), a location server, and/or a location management function (LMF) according to the embodiment.

[0082] FIG. **13** is a simplified diagram illustrating a method of operating a UE, a TRP, a location server, and/or an LMF according to the embodiment.

[0083] FIG. **14** is a diagram illustrating an example of a method of configuring a BW according to one embodiment.

[0084] FIG. **15** is a flowchart illustrating a method for operating a UE and a network node according to one embodiment.

[0085] FIG. **16** is a flowchart illustrating a method for operating a UE according to one embodiment.

[0086] FIG. **17** is a flowchart illustrating a method for operating a network node according to one embodiment.

[0087] FIG. **18** is a diagram illustrating devices that implement an embodiment of the disclosure.

[0088] FIG. **19** illustrates an exemplary communication system applied to an embodiment of the disclosure.

[0089] FIG. **20** illustrates exemplary wireless devices applied to an embodiment.

[0090] FIG. **21** illustrates other exemplary wireless devices applied to an embodiment.

[0091] FIG. **22** illustrates an exemplary portable device applied to an embodiment.

[0092] FIG. **23** illustrates an exemplary vehicle or autonomous driving vehicle applied to an embodiment.

DETAILED DESCRIPTION

[0093] The embodiment is applicable to a variety of wireless access technologies such as code division multiple access (CDMA), frequency division multiple access (FDMA), time division multiple access (TDMA), orthogonal frequency division multiple access (OFDMA), and single carrier frequency division multiple access (SC-FDMA). CDMA can be implemented as a radio technology such as Universal Terrestrial Radio Access (UTRA) or CDMA2000. TDMA can be implemented as a radio technology such as Global System for Mobile communications (GSM)/General Packet Radio Service (GPRS)/Enhanced Data Rates for GSM Evolution (EDGE). OFDMA can be implemented as a radio technology such as Institute of Electrical and Electronics Engineers (IEEE) 802.11 (Wireless Fidelity (Wi-Fi)), IEEE 802.16 (Worldwide interoperability for Microwave Access (WiMAX)), IEEE 802.20, and Evolved UTRA (E-UTRA). UTRA is a part of Universal Mobile Telecommunications System (UMTS). 3rd Generation Partnership Project (3GPP) Long Term Evolution (LTE) is part of Evolved UMTS (E-UMTS) using E-UTRA, and LTE-Advanced (A) is an evolved version of 3GPP LTE. 3GPP NR (New Radio or New Radio Access Technology) is an evolved version of 3GPP LTE/LTE-A.

[0094] The embodiment is described in the context of a 3GPP communication system (e.g., including LTE, NR, 6G, and next-generation wireless communication systems) for clarity of description, to which the technical spirit of the embodiment is not limited. For the background art, terms, and abbreviations used in the description of the embodiment, refer to the technical specifications published before the present disclosure. For example, the documents of 3GPP TS 36.211, 3GPP TS 36.212, 3GPP TS 36.213, 3GPP TS 36.300, 3GPP TS 36.321, 3GPP TS 36.331, 3GPP TS 36.355, 3GPP TS 36.455, 3GPP TS 37.355, 3GPP TS 37.455, 3GPP TS 38.211, 3GPP TS 38.212, 3GPP TS 38.213, 3GPP TS 38.214, 3GPP TS 38.215, 3GPP TS 38.300, 3GPP TS 38.321, 3GPP TS 38.331, 3GPP TS 38.355, 3GPP TS 38.455, and so on may be referred to.'

1. 3GPP System

1.1. Physical Channels and Signal Transmission and Reception

[0095] In a wireless access system, a UE receives information from a base station on a downlink (DL) and transmits information to the base station on an uplink (UL). The information transmitted and received between the UE and the base station includes general data information and various types of control information. There are many physical channels according to the types/usages of information transmitted and received between the base station and the UE.

[0096] FIG. **1** is a diagram illustrating physical channels and a signal transmission method using the physical channels, which may be used in the embodiment.

[0097] When powered on or when a UE initially enters a cell, the UE performs initial cell search involving synchronization with a BS in step S**11**. For initial cell search, the UE receives a synchronization signal block (SSB). The SSB includes a primary synchronization signal (PSS), a secondary synchronization signal (SSS), and a physical broadcast channel (PBCH). The UE synchronizes with the BS and acquires information such as a cell Identifier (ID) based on the PSS/SSS. Then the UE may receive broadcast information from the cell on the PBCH. In the meantime, the UE may check a downlink channel status by receiving a downlink reference signal (DL RS) during initial cell search.

[0098] After initial cell search, the UE may acquire more specific system information by receiving a physical downlink control channel (PDCCH) and receiving a physical downlink shared channel (PDSCH) based on information of the PDCCH in step S**12**.

[0099] Subsequently, to complete connection to the eNB, the UE may perform a random access procedure with the eNB (S**13** to S**16**). In the random access procedure, the UE may transmit a preamble on a physical random access channel (PRACH) (S**13**) and may receive a PDCCH and a random access response (RAR) for the preamble on a PDSCH associated with the PDCCH (S**14**). The UE may transmit a physical uplink shared channel (PUSCH) by using scheduling information in the RAR (S**15**), and perform a contention resolution procedure including reception of a PDCCH signal and a PDSCH signal corresponding to the PDCCH signal (S**16**).

[0100] Aside from the above 4-step random access procedure (4-step RACH procedure or type-1 random access procedure), when the random access procedure is performed in two steps (2-step RACH procedure or type-2 random access procedure), steps S**13** and S**15** may be performed as one UE transmission operation (e.g., an operation of transmitting message A (MsgA) including a PRACH preamble and/or a PUSCH), and steps S**14** and S**16** may be performed as one BS transmission operation (e.g., an operation of transmitting message B (MsgB) including an RAR and/or contention resolution information)

[0101] After the above procedure, the UE may receive a PDCCH and/or a PDSCH from the BS (S**17**) and transmit a PUSCH and/or a physical uplink control channel (PUCCH) to the BS (S**18**), in a general UL/DL signal transmission procedure.

[0102] Control information that the UE transmits to the BS is generically called uplink control information (UCI). The UCI includes a hybrid automatic repeat and request acknowledgement/negative acknowledgement (HARQ-ACK/NACK), a scheduling request (SR), a channel quality indicator (CQI), a precoding matrix index (PMI), a rank indicator (RI), etc.

[0103] In general, UCI is transmitted periodically on a PUCCH. However, if control information and traffic data should be transmitted simultaneously, the control information and traffic data may be transmitted on a PUSCH. In addition, the UCI may be transmitted aperiodically on the PUSCH, upon receipt of a request/command from a network.

1.2. Physical Resource

[0104] FIG. **2** is a diagram illustrating a radio frame structure in a new radio access technology (NR) system to which the embodiment is applicable.

[0105] The NR system may support multiple numerologies. A numerology may be defined by a subcarrier spacing (SCS) and a cyclic prefix (CP) overhead. Multiple SCSs may be derived by scaling a default SCS by an integer N (or u). Further, even though it is assumed that a very small SCS is not used in a very high carrier frequency, a numerology to be used may be selected independently of the frequency band of a cell. Further, the NR system may support various frame structures according to multiple numerologies.

[0106] Now, a description will be given of OFDM numerologies and frame structures which may be considered for the NR system. Multiple OFDM numerologies supported by the NR system may be defined as listed in Table 1. For a bandwidth part, u and a CP are obtained from RRC parameters provided by the BS.

TABLE 1

μ	$\Delta f = 2^{\mu} \cdot 15$[kHz]	Cyclic prefix
0	15	Normal
1	30	Normal
2	60	Normal, Extended
3	120	Normal
4	240	Normal

[0107] In NR, multiple numerologies (e.g., SCSs) are supported to support a variety of 5G services. For example, a wide area in cellular bands is supported for an SCS of 15 kHz, a dense-urban area, a lower latency, and a wider carrier bandwidth are supported for an SCS of 30/60 kHz, and a larger bandwidth than 24.25 GHz is supported for an SCS of 60 kHz or more, to overcome phase noise.

[0108] An NR frequency band is defined by two types of frequency ranges, FR1 and FR2. FR1 may be a sub-6 GHz range, and FR2 may be an above-6 GHz range, that is, a millimeter wave (mmWave) band.

[0109] Table 2 below defines the NR frequency band, by way of example.

TABLE 2

Frequency range designation	Corresponding frequency range	Subcarrier Spacing
FR1	410 MHz-7125 MHz	15, 30, 60 kHz
FR2	24250 MHz-52600 MHz	60, 120, 240 kHz

[0110] Regarding a frame structure in the NR system, the time-domain sizes of various fields are represented as multiples of a basic time unit for NR, Tc=1/(Δfmax*Nf) where Δfmax=480*103 Hz and a value Nf related to a fast Fourier transform (FFT) size or an inverse fast Fourier transform (IFFT) size is given as Nf=4096. Tc and Ts which is an LTE-based time unit and sampling time, given as Ts=1/((15 kHz)*2048) are placed in the following relationship: Ts/Tc=64. DL and UL transmissions are organized into (radio) frames each having a duration of Tf=(Δfmax*Nf/100)*Tc=10 ms. Each radio frame includes 10 subframes each having a duration of Tsf=(Δfmax*Nf/1000)*Tc=1 ms. There may exist one set of frames for UL and one set of frames for DL. For a numerology u, slots are numbered with nμs∈{0, . . . ,Nslot,μsubframe−1} in an increasing order in a subframe, and with nμs,f∈ {0, . . . ,Nslot,μframe−1} in an increasing order in a radio frame. One slot includes Nusymb consecutive OFDM symbols, and Nμsymb depends on a CP. The start of a slot nus in a subframe is aligned in time with the start of an OFDM symbol nμs*Nμsymb in the same subframe.

[0111] Table 3 lists the number of symbols per slot, the number of slots per frame, and the number of slots per subframe, for each SCS in a normal CP case, and Table 4 lists the number of symbols per slot, the number of slots per frame, and the number of slots per subframe, for each SCS in an extended CP case.

TABLE 3

μ	N_{symb}^{slot}	$N_{slot}^{frame,\mu}$	$N_{slot}^{subframe,\mu}$
0	14	10	1
1	14	20	2
2	14	40	4
3	14	80	8
4	14	160	16

TABLE 4

μ	N_{symb}^{slot}	$N_{slot}^{frame,\mu}$	$N_{slot}^{subframe,\mu}$
2	12	40	4

[0112] In the above tables, $N^{slot}{}_{symb}$ represents the number of symbols in a slot, $N^{frame,\mu}{}_{slot}$ represents the number of slots in a frame, and $N_{subframe,\mu}{}^{slot}$ represents the number of slots in a subframe.'

[0113] In the NR system to which the embodiment is applicable, different OFDM (A) numerologies (e.g., SCSs, CP lengths, and so on) may be configured for a plurality of cells which are aggregated for one UE. Accordingly, the (absolute time) period of a time resource including the same number of symbols (e.g., a subframe (SF), a slot, or a TTI) (generically referred to as a time unit (TU), for convenience) may be configured differently for the aggregated cells.

[0114] FIG. **2** illustrates an example with μ=2 (i.e., an SCS of 60 kHz), in which referring to Table 6, one subframe may include four slots. One subframe={1, 2, 4} slots in FIG. **7**, which is exemplary, and the number of slot(s) which may be included in one subframe is defined as listed in Table 3 or Table 4.

[0115] Further, a mini-slot may include 2, 4 or 7 symbols, fewerBWols than 2, or more symbols than 7.

[0116] Regarding physical resources in the NR system, antenna ports, a resource grid, resource elements (REs), resource blocks (RBs), carrier parts, and so one may be considered. The physical resources in the NR system will be described below in detail.

[0117] An antenna port is defined such that a channel conveying a symbol on an antenna port may be inferred from a channel conveying anotherBWol on the same antenna port. When the large-scale properties of a channel carrying a symbol on one antenna port may be inferred from a channel carrying a symbol on another antenna port, the two antenna ports may be said to be in a quasi-co-located or quasi-co-location (QCL) relationship. The large-scale properties include one or more of delay spread, Doppler spread, frequency shift, average received power, received timing, average delay, and a spatial reception (Rx) parameter. The spatial Rx parameter refers to a spatial (Rx) channel property parameter such as an angle of arrival.

[0118] FIG. **3** illustrates an exemplary resource grid to which the embodiment is applicable.

[0119] Referring to FIG. **3**, for each subcarrier spacing (SCS) and carrier, a resource grid is defined as $14\times2^{\mu}$ OFDM symbols by $N_{grid}^{size,\mu}\times N_{SC}^{RB}$ subcarriers, where $N_{grid}^{size,\mu}$ is indicated by RRC signaling from the BS. $N_{grid}^{size,\mu}$ may vary according to an SCS configuration u and a transmission direction, UL or DL. There is one resource grid for an SCS configuration u, an antenna port p, and a transmission direction (UL or DL). Each element of the resource grid for the SCS configuration μ and the antenna port p is referred to as an RE and uniquely identified by an index pair (k, l) where k represents an index in the frequency domain, and l represents a symbol position in the frequency domain relative to a reference point. The RE (k, l) for the SCS configuration u and the antenna port p corresponds to a physical resource and a complex value $a_{k,l}^{(p,\mu)}$. An RB is defined as N_{SC}^{RB}=12 consecutive subcarriers in the frequency domain.

[0120] Considering that the UE may not be capable of supporting a wide bandwidth supported in the NR system, the UE may be configured to operate in a part (bandwidth part (BWP)) of the frequency bandwidth of a cell.

[0121] FIG. **4** is a diagram illustrating exemplary mapping of physical channels in a slot, to which the embodiment is applicable.

[0122] One slot may include all of a DL control channel, DL or UL data, and a UL control channel. For example, the first N symbols of a slot may be used to transmit a DL control channel (hereinafter, referred to as a DL control region), and the last M symbols of the slot may be used to transmit a UL control channel (hereinafter, referred to as a UL control region). Each of N and M is an integer equal to or larger than **0**. A resource area (hereinafter, referred to as a data region) between the DL control region and the UL control region may be used to transmit DL data or UL data. There may be a time gap for DL-to-UL or UL-to-DL switching between a control region and a data region. A PDCCH may be transmitted in the DL control region, and a PDSCH may be transmitted in the DL data region. Some symbols at a DL-to-UL switching time in the slot may be used as the time gap.

[0123] The BS transmits related signals to the UE on DL channels as described below, and the UE receives the related signals from the BS on the DL channels.

[0124] The PDSCH conveys DL data (e.g., DL-shared channel transport block (DL-SCH TB)) and uses a modulation scheme such as quadrature phase shift keying (QPSK), 16-ary quadrature amplitude modulation (16QAM), 64QAM, or 256QAM. A TB is encoded into a codeword. The PDSCH may deliver up to two codewords. Scrambling and modulation mapping are performed on a codeword basis, and modulation symbols generated from each codeword are mapped to one or more layers (layer mapping). Each layer together with a demodulation reference signal (DMRS) is mapped to resources, generated as an OFDM symbol signal, and transmitted through a corresponding antenna port.

[0125] The PDCCH may deliver downlink control information (DCI), for example, DL data scheduling information, UL data scheduling information, and so on. The PUCCH may deliver uplink control information (UCI), for example, an acknowledgement/negative acknowledgement (ACK/NACK) information for DL data, channel state information (CSI), a scheduling request (SR), and so on.

[0126] The PDCCH carries downlink control information (DCI) and is modulated in quadrature phase shift keying (QPSK). One PDCCH includes 1, 2, 4, 8, or 16 control channel elements (CCEs) according to an aggregation level (AL). One CCE includes 6 resource element groups (REGs). One REG is defined by one OFDM symbol by one (P) RB.

[0127] The PDCCH is transmitted in a control resource set (CORESET). A CORESET is defined as a set of REGs having a given numerology (e.g., SCS, CP length, and so on). A plurality of CORESETs for one UE may overlap with each other in the time/frequency domain. A CORESET may be configured by system information (e.g., a master information block (MIB)) or by UE-specific higher layer (RRC) signaling. Specifically, the number of RBs and the number of symbols (up to 3 symbols) included in a CORESET may be configured by higher-layer signaling.

[0128] The UE acquires DCI delivered on a PDCCH by decoding (so-called blind decoding) a set of PDCCH candidates. A set of PDCCH candidates decoded by a UE are defined as a PDCCH search space set. A search space set may be a common search space (CSS) or a UE-specific search space (USS). The UE may acquire DCI by monitoring PDCCH candidates in one or more search space sets configured by an MIB or higher-layer signaling.

[0129] The UE transmits related signals on later-described UL channels to the BS, and the BS receives the related signals on the UL channels from the UE.

[0130] The PUSCH delivers UL data (e.g., a UL-shared channel transport block (UL-SCH TB)) and/or UCI, in cyclic prefix-orthogonal frequency division multiplexing (CP-OFDM) waveforms or discrete Fourier transform-spread-orthogonal division multiplexing (DFT-s-OFDM) waveforms. If the PUSCH is transmitted in DFT-s-OFDM waveforms, the UE transmits the PUSCH by applying transform precoding. For example, if transform precoding is impossible (e.g., transform precoding is disabled), the UE may transmit the PUSCH in CP-OFDM waveforms, and if transform precoding is possible (e.g., transform precoding is enabled), the UE may transmit the PUSCH in CP-OFDM waveforms or DFT-s-OFDM waveforms. The PUSCH transmission may be scheduled dynamically by a UL grant in DCI or semi-statically by higher-layer signaling (e.g., RRC signaling) (and/or layer **1** (L**1**) signaling (e.g., a PDCCH)) (a configured grant). The PUSCH transmission may be performed in a codebook-based or non-codebook-based manner.

[0131] The PUCCH delivers UCI, an HARQ-ACK, and/or an SR and is classified as a short PUCCH or a long PUCCH according to the transmission duration of the PUCCH.

2. Positioning

[0132] Positioning may refer to determining the geographical position and/or velocity of the UE based on measurement of radio signals. Location information may be requested by and reported to a client (e.g., an application) associated with to the UE. The location information may also be requested by a client within or connected to a core network. The location information may be reported in standard formats such as formats for cell-based or geographical coordinates, together with estimated errors of the position and velocity of the UE and/or a positioning method used for positioning.

2.1. Positioning Protocol Configuration

[0133] FIG. **5** is a diagram illustrating an exemplary positioning protocol configuration for positioning a UE, to which the embodiment is applicable.

[0134] Referring to FIG. **5**, an LTE positioning protocol (LPP) may be used as a point-to-point protocol between a location server (E-SMLC and/or SLP and/or LMF) and a target device (UE and/or SET), for positioning the target device using position-related measurements obtained from one or more reference resources. The target device and the location server may exchange measurements and/or location information based on signal A and/or signal B over the LPP.

[0135] NRPPa may be used for information exchange between a reference source (access node and/or BS and/or TP and/or NG-RAN node) and the location server.

[0136] The NRPPa protocol may provide the following functions.

- **[0137]** E-CID Location Information Transfer. This function allows the reference source to exchange location information with the LMF for the purpose of E-CID positioning.
- **[0138]** OTDOA Information Transfer. This function allows the reference source to exchange information with the LMF for the purpose of OTDOA positioning.
- **[0139]** Reporting of General Error Situations. This function allows reporting of general error situations, for which function-specific error messages have not been defined.

2.2. PRS (Positioning Reference Signal)

[0140] For such positioning, a positioning reference signal (PRS) may be used. The PRS is a reference signal used to estimate the position of the UE.

[0141] A positioning frequency layer may include one or more PRS resource sets, each including one or more PRS resources.

Sequence Generation

[0142] A PRS sequence r(m) (m=0,1, . . .) may be defined by Equation 1.

$$r(m) = \frac{1}{\sqrt{2}}(1 - 2c(m)) + j\frac{1}{\sqrt{2}}(1 - 2c(m+1)) \qquad \text{[Equation 1]}$$

[0143] In Equation 1, c(i) may be a pseudo-random sequence. A pseudo-random sequence generator may be initialized by Equation 2.

$$c_{init} = \left(2^{22}\left\lfloor\frac{n_{ID,seq}^{PRS}}{1024}\right\rfloor + 2^{10}\left(N_{symb}^{slot}n_{s,f}^{\mu} + l + 1\right)\left(2\left(n_{ID,seq}^{PRS} \bmod 1024\right) + 1\right) + \left(n_{ID,seq}^{PRS} \bmod 1024\right)\right) \bmod 2^{31} \qquad \text{[Equation 2]}$$

[0144] $n_{s,f}^{\mu}$ may be a slot number in a frame in an SCS configuration μ. A DL PRS sequence ID $n_{ID,seq}^{PRS} \in \{0,1, \ldots ,4095\}$ may be given by a higher-layer parameter (e.g., DL-PRS-SequenceId). l may be an OFDM symbol in a slot to which the sequence is mapped.

Mapping to Physical Resources in a DL PRS Resource A PRS sequence r(m) may by scaled by β_{PRS} and mapped to REs $(k,l)_{p,\mu}$, specifically by Equation 3. $(k, l)_{p,\mu}$ may represent an RE (k, l) for an antenna port p and the SCS configuration μ.

$$a_{k,l}^{(p,\mu)} = \beta_{PRS}r(m) \qquad \text{[Equation 3]}$$

$$m = 0, 1, \ldots$$

$$k = mK_{comb}^{PRS} + \left(\left(k_{offset}^{PRS} + k'\right) \bmod K_{comb}^{PR}\right)$$

$$l = l_{start}^{PRS}, l_{start}^{PRS} + 1, \ldots , l_{start}^{PRS} + L_{PRS} - 1$$

[0145] Herein, the following conditions may have to be satisfied:

- **[0146]** The REs $(k,l)_{p,\mu}$ are included in an RB occupied by a DL PRS resource configured for the UE;
- **[0147]** The symbol l not used by any SS/PBCH block used by a serving cell for a DL PRS transmitted from the serving cell or indicated by a higher-layer parameter SSB-positionInBurst for a DL PRS transmitted from a non-serving cell;
- **[0148]** A slot number satisfies the following PRS resource set-related condition;

[0149] l_{start}^{PRS} is the first symbol of the DL PRS in the slot, which may be given by a higher-layer parameter DL-PRS-ResourceSymbolOffset. The time-domain size of the DL PRS resource, $L_{PRS} \in \{2,4,6,12\}$ may be given by a higher-layer parameter DL-PRS-NumSymbols. A comb size comb $K_{comb}^{PRS} \in \{2, 4, 6,12\}$ may be given by a higher-layer parameter transmissionComb. A combination $\{L_{PRS}, K_{comb}^{PRS}\}$ of L_{PRS} and K_{comb}^{PRS} may be one of {2, 2}, {4, 2}, {6, 2}, {12, 2}, {4, 4}, {12, 4}, {6, 6}, {12, 6} and/or {12, 12}. An RE offset $K_{offset}^{PRS} \in \{0,1, \ldots , K_{comb}^{PRS}-1\}$ may be given by combOffset. A frequency offset k' may be a function of $l-l_{start}^{PRS}$ as shown in Table 5.

TABLE 5

	Symbol number within the downlink PRS resource $l - l_{start}^{PRS}$											
K_{comb}^{PRS}	0	1	2	3	4	5	6	7	8	9	10	11
2	0	1	0	1	0	1	0	1	0	1	0	1
4	0	2	1	3	0	2	1	3	0	2	1	3
6	0	3	1	4	2	5	0	3	1	4	2	5
12	0	6	3	9	1	7	4	10	2	8	5	11

[0150] A reference point for k=0 may be the position of point A in a positioning frequency layer in which the DL PRS resource is configured. Point A may be given by a higher-layer parameter dl-PRS-PointA-r16.

Mapping to Slots in a DL PRS Resource Set

[0151] A DL PRS resource included in a DL PRS resource set may be transmitted in a slot and a frame which satisfy the following Equation 4.

$$\left(N_{slot}^{frame,\mu}n_f + n_{s,f}^{\mu} - T_{offset}^{PRS} - T_{offset,res}^{PRS}\right) \bmod 2^{\mu}T^{PRS} \in \left\{iT_{gap}^{PRS}\right\}_{i=0}^{T_{rep}^{PRS}-1} \qquad \text{[Equation 4]}$$

[0152] $N_{slot}^{frame,\mu}$ may be the number of slots per frame in the SCS configuration μ. n_f may be a system frame number (SFN). $n_{s,f}^{\mu}$ may be a slot number in a frame in the SCS configuration μ. A slot offset $T_{offset}^{PRS} \in \{0,1, \ldots , T_{per}^{PRS}-1\}$ may be given by a higher-layer parameter DL-PRS-ResourceSetSlotOffset. A DL PRS resource slot offset $T_{offset,}$

$_{res}^{PRS}$ may be given by a higher layer parameter DL-PRS-ResourceSlotOffset. A periodicity $T_{per}^{PRS} \in \{4, 5, 8, 10, 16, 20, 32, 40, 64, 80, 160, 320, 640, 1280, 2560, 5120, 10240\}$ may be given by a higher-layer parameter DL-PRS-Periodicity. A repetition factor $T_{rep}^{PRS} \in = \{1,2,4,6,8,16,32\}$ may be given by a higher-layer parameter DL-PRS-ResourceRepetitionFactor. A muting repetition factor T_{muting}^{PRS} may be given by a higher-layer parameter DL-PRS-MutingBitRepetitionFactor. A time gap $T_{gap}^{PRS} \in \{1,2,4,8,16,32\}$ may be given by a higher-layer parameter DL-PRS-ResourceTimeGap.

2.3. UE Positioning Architecture

[0153] FIG. **6** illustrates an exemplary system architecture for measuring positioning of a UE to which the embodiment is applicable.

[0154] Referring to FIG. **6**, an AMF may receive a request for a location service associated with a particular target UE from another entity such as a gateway mobile location center (GMLC) or the AMF itself decides to initiate the location service on behalf of the particular target UE. Then, the AMF transmits a request for a location service to a location management function (LMF). Upon receiving the request for the location service, the LMF may process the request for the location service and then returns the processing result including the estimated position of the UE to the AMF. In the case of a location service requested by an entity such as the GMLC other than the AMF, the AMF may transmit the processing result received from the LMF to this entity.

[0155] A new generation evolved-NB (ng-eNB) and a gNB are network elements of the NG-RAN capable of providing a measurement result for positioning. The ng-eNB and the gNB may measure radio signals for a target UE and transmits a measurement result value to the LMF. The ng-eNB may control several TPs, such as remote radio heads, or PRS-only TPs for support of a PRS-based beacon system for E-UTRA.

[0156] The LMF is connected to an enhanced serving mobile location center (E-SMLC) which may enable the LMF to access the E-UTRAN. For example, the E-SMLC may enable the LMF to support OTDOA, which is one of positioning methods of the E-UTRAN, using DL measurement obtained by a target UE through signals transmitted by eNBs and/or PRS-only TPs in the E-UTRAN.

[0157] The LMF may be connected to an SUPL location platform (SLP). The LMF may support and manage different location services for target UEs. The LMF may interact with a serving ng-eNB or a serving gNB for a target UE in order to obtain position measurement for the UE. For positioning of the target UE, the LMF may determine positioning methods, based on a location service (LCS) client type, required quality of service (QOS), UE positioning capabilities, gNB positioning capabilities, and ng-eNB positioning capabilities, and then apply these positioning methods to the serving gNB and/or serving ng-eNB. The LMF may determine additional information such as accuracy of the location estimate and velocity of the target UE. The SLP is a secure user plane location (SUPL) entity responsible for positioning over a user plane.

[0158] The UE may measure the position thereof using DL RSs transmitted by the NG-RAN and the E-UTRAN. The DL RSs transmitted by the NG-RAN and the E-UTRAN to the UE may include a SS/PBCH block, a CSI-RS, and/or a PRS. Which DL RS is used to measure the position of the UE may conform to configuration of LMF/E-SMLC/ng-eNB/E-UTRAN etc. The position of the UE may be measured by an RAT-independent scheme using different global navigation satellite systems (GNSSs), terrestrial beacon systems (TBSs), WLAN access points, Bluetooth beacons, and sensors (e.g., barometric sensors) installed in the UE. The UE may also contain LCS applications or access an LCS application through communication with a network accessed thereby or through another application contained therein. The LCS application may include measurement and calculation functions needed to determine the position of the UE. For example, the UE may contain an independent positioning function such as a global positioning system (GPS) and report the position thereof, independent of NG-RAN transmission. Such independently obtained positioning information may be used as assistance information of positioning information obtained from the network.

2.4. Operation for UE Positioning

[0159] FIG. **7** illustrates an implementation example of a network for UE positioning.

[0160] When an AMF receives a request for a location service in the case in which the UE is in connection management (CM)-IDLE state, the AMF may make a request for a network triggered service in order to establish a signaling connection with the UE and to assign a specific serving gNB or ng-eNB. This operation procedure is omitted in FIG. **7**. In other words, in FIG. **7** it may be assumed that the UE is in a connected mode. However, the signaling connection may be released by an NG-RAN as a result of signaling and data inactivity while a positioning procedure is still ongoing.

[0161] An operation procedure of the network for UE positioning will now be described in detail with reference to FIG. **7**. In step **1***a*, a 5GC entity such as GMLC may transmit a request for a location service for measuring the position of a target UE to a serving AMF. Here, even when the GMLC does not make the request for the location service, the serving AMF may determine the need for the location service for measuring the position of the target UE according to step **1***b*. For example, the serving AMF may determine that itself will perform the location service in order to measure the position of the UE for an emergency call.

[0162] In step **2**, the AMF transfers the request for the location service to an LMF. In step **3***a*, the LMF may initiate location procedures with a serving ng-eNB or a serving gNB to obtain location measurement data or location measurement assistance data. For example, the LMF may transmit a request for location related information associated with one or more UEs to the NG-RAN and indicate the type of necessary location information and associated QoS. Then, the NG-RAN may transfer the location related information to the LMF in response to the request. In this case, when a location determination method according to the request is an enhanced cell ID (E-CID) scheme, the NG-RAN may transfer additional location related information to the LMF in one or more NR positioning protocol A (NRPPa) messages. Here, the "location related information" may mean all values used for location calculation such as actual location estimate information and radio measurement or location measurement. Protocol used in step **3***a* may be an NRPPa protocol which will be described later.

[0163] Additionally, in step **3***b*, the LMF may initiate a location procedure for DL positioning together with the UE.

For example, the LMF may transmit the location assistance data to the UE or obtain a location estimate or location measurement value. For example, in step **3***b*, a capability information transfer procedure may be performed. Specifically, the LMF may transmit a request for capability information to the UE and the UE may transmit the capability information to the LMF. Here, the capability information may include information about a positioning method supportable by the LFM or the UE, information about various aspects of a particular positioning method, such as various types of assistance data for an A-GNSS, and information about common features not specific to any one positioning method, such as ability to handle multiple LPP transactions. In some cases, the UE may provide the capability information to the LMF although the LMF does not transmit a request for the capability information.

[0164] As another example, in step **3***b*, a location assistance data transfer procedure may be performed. Specifically, the UE may transmit a request for the location assistance data to the LMF and indicate particular location assistance data needed to the LMF. Then, the LMF may transfer corresponding location assistance data to the UE and transfer additional assistance data to the UE in one or more additional LTE positioning protocol (LPP) messages. The location assistance data delivered from the LMF to the UE may be transmitted in a unicast manner. In some cases, the LMF may transfer the location assistance data and/or the additional assistance data to the UE without receiving a request for the assistance data from the UE.

[0165] As another example, in step **3***b*, a location information transfer procedure may be performed. Specifically, the LMF may send a request for the location (related) information associated with the UE to the UE and indicate the type of necessary location information and associated QoS. In response to the request, the UE may transfer the location related information to the LMF. Additionally, the UE may transfer additional location related information to the LMF in one or more LPP messages. Here, the "location related information" may mean all values used for location calculation such as actual location estimate information and radio measurement or location measurement. Typically, the location related information may be a reference signal time difference (RSTD) value measured by the UE based on DL RSs transmitted to the UE by a plurality of NG-RANs and/or E-UTRANs. Similarly to the above description, the UE may transfer the location related information to the LMF without receiving a request from the LMF.

[0166] The procedures implemented in step **3***b* may be performed independently but may be performed consecutively. Generally, although step **3***b* is performed in order of the capability information transfer procedure, the location assistance data transfer procedure, and the location information transfer procedure, step **3***b* is not limited to such order. In other words, step **3***b* is not required to occur in specific order in order to improve flexibility in positioning. For example, the UE may request the location assistance data at any time in order to perform a previous request for location measurement made by the LMF. The LMF may also request location information, such as a location measurement value or a location estimate value, at any time, in the case in which location information transmitted by the UE does not satisfy required QoS. Similarly, when the UE does not perform measurement for location estimation, the UE may transmit the capability information to the LMF at any time.

[0167] In step **3***b*, when information or requests exchanged between the LMF and the UE are erroneous, an error message may be transmitted and received and an abort message for aborting positioning may be transmitted and received.

[0168] Protocol used in step **3***b* may be an LPP protocol which will be described later.

[0169] Step **3***b* may be performed additionally after step **3***a* but may be performed instead of step **3***a*.

[0170] In step **4**, the LMF may provide a location service response to the AMF. The location service response may include information as to whether UE positioning is successful and include a location estimate value of the UE. If the procedure of FIG. **7** has been initiated by step **1***a*, the AMF may transfer the location service response to a 5GC entity such as a GMLC.

[0171] If the procedure of FIG. **9** has been initiated by step **1***b*, the AMF may use the location service response in order to provide a location service related to an emergency call.

2.5. Positioning Protocol

LTE Positioning Protocol (LPP)

[0172] FIG. **8** illustrates an exemplary protocol layer used to support LPP message transfer between an LMF and a UE. An LPP protocol data unit (PDU) may be carried in a NAS PDU between an AMF and the UE.

[0173] Referring to FIG. **10**, LPP is terminated between a target device (e.g., a UE in a control plane or an SUPL enabled terminal (SET) in a user plane) and a location server (e.g., an LMF in the control plane or an SLP in the user plane). LPP messages may be carried as transparent PDUs cross intermediate network interfaces using appropriate protocols, such an NGAP over an NG-C interface and NAS/RRC over LTE-Uu and NR-Uu interfaces. LPP is intended to enable positioning for NR and LTE using various positioning methods.

[0174] For example, a target device and a location server may exchange, through LPP, capability information therebetween, assistance data for positioning, and/or location information. The target device and the location server may exchange error information and/or indicate abort of an LPP procedure, through an LPP message.

NR Positioning Protocol A (NRPPa)

[0175] FIG. **9** illustrates an exemplary protocol layer used to support NRPPa PDU transfer between an LMF and an NG-RAN node.

[0176] NRPPa may be used to carry information between an NG-RAN node and an LMF. Specifically, NRPPa may carry an E-CID for measurement transferred from an ng-eNB to an LMF, data for support of an OTDOA positioning method, and a cell-ID and a cell position ID for support of an NR cell ID positioning method. An AMF may route NRPPa PDUs based on a routing ID of an involved LMF over an NG-C interface without information about related NRPPa transaction.

[0177] An NRPPa procedure for location and data collection may be divided into two types. The first type is a UE associated procedure for transfer of information about a particular UE (e.g., location measurement information) and the second type is a non-UE-associated procedure for transfer of information applicable to an NG-RAN node and associated TPs (e.g., gNB/ng-eNB/TP timing information). The two types may be supported independently or may be supported simultaneously.

2.6. Positioning Measurement Method

[0178] Positioning methods supported in the NG-RAN may include a GNSS, an OTDOA, an E-CID, barometric sensor positioning, WLAN positioning, Bluetooth positioning, a TBS, uplink time difference of arrival (UTDOA) etc. Although any one of the positioning methods may be used for UE positioning, two or more positioning methods may be used for UE positioning.

OTDOA (Observed Time Difference Of Arrival)

[0179] FIG. **10** is a diagram illustrating an observed time difference of arrival (OTDOA) positioning method, to which the embodiment is applicable;

[0180] The OTDOA positioning method uses time measured for DL signals received from multiple TPs including an eNB, an ng-eNB, and a PRS-only TP by the UE. The UE measures time of received DL signals using location assistance data received from a location server. The position of the UE may be determined based on such a measurement result and geographical coordinates of neighboring TPs.

[0181] The UE connected to the gNB may request measurement gaps to perform OTDOA measurement from a TP. If the UE is not aware of an SFN of at least one TP in OTDOA assistance data, the UE may use autonomous gaps to obtain an SFN of an OTDOA reference cell prior to requesting measurement gaps for performing reference signal time difference (RSTD) measurement.

[0182] Here, the RSTD may be defined as the smallest relative time difference between two subframe boundaries received from a reference cell and a measurement cell. That is, the RSTD may be calculated as the relative time difference between the start time of a subframe received from the measurement cell and the start time of a subframe from the reference cell that is closest to the subframe received from the measurement cell. The reference cell may be selected by the UE.

[0183] For accurate OTDOA measurement, it is necessary to measure time of arrival (ToA) of signals received from geographically distributed three or more TPs or BSs. For example, ToA for each of TP **1**, TP **2**, and TP **3** may be measured, and RSTD for TP **1** and TP **2**, RSTD for TP **2** and TP **3**, and RSTD for TP **3** and TP **1** are calculated based on three ToA values. A geometric hyperbola is determined based on the calculated RSTD values and a point at which curves of the hyperbola cross may be estimated as the position of the UE. In this case, accuracy and/or uncertainty for each ToA measurement may occur and the estimated position of the UE may be known as a specific range according to measurement uncertainty.

[0184] For example, RSTD for two TPs may be calculated based on Equation 5 below.

[Equation 5]

$$RSTDi_{,1} = \frac{\sqrt{(x_t - x_i)^2 + (y_t - y_i)^2}}{c} -$$

-continued

$$\frac{\sqrt{(x_t - x_1)^2 + (y_t - y_1)^2}}{c} + (T_i - T_1) + (n_i - n_1)$$

[0185] In Equation 5, c is the speed of light, $\{x_t, y_t\}$ are (unknown) coordinates of a target UE, $\{x_i, y_i\}$ are (known) coordinates of a TP, and $\{x_1, y_1\}$ are coordinates of a reference TP (or another TP). Here, $(T_i\text{-}T_1)$ is a transmission time offset between two TPs, referred to as "real time differences" (RTDs), and n_i and n_1 are UE ToA measurement error values.

E-CID (Enhanced Cell ID)

[0186] In a cell ID (CID) positioning method, the position of the UE may be measured based on geographical information of a serving ng-eNB, a serving gNB, and/or a serving cell of the UE. For example, the geographical information of the serving ng-eNB, the serving gNB, and/or the serving cell may be acquired by paging, registration, etc.

[0187] The E-CID positioning method may use additional UE measurement and/or NG-RAN radio resources in order to improve UE location estimation in addition to the CID positioning method. Although the E-CID positioning method partially may utilize the same measurement methods as a measurement control system on an RRC protocol, additional measurement only for UE location measurement is not generally performed. In other words, an additional measurement configuration or measurement control message may not be provided for UE location measurement. The UE does not expect that an additional measurement operation only for location measurement will be requested and the UE may report a measurement value obtained by generally measurable methods.

[0188] For example, the serving gNB may implement the E-CID positioning method using an E-UTRA measurement value provided by the UE.

[0189] Measurement elements usable for E-CID positioning may be, for example, as follows.

- **[0190]** UE measurement: E-UTRA reference signal received power (RSRP), E-UTRA reference signal received quality (RSRQ), UE E-UTRA reception (Rx)-transmission (Tx) time difference, GERAN/WLAN reference signal strength indication (RSSI), UTRAN common pilot channel (CPICH) received signal code power (RSCP), and/or UTRAN CPICH Ec/Io
- **[0191]** E-UTRAN measurement: ng-eNB Rx-Tx time difference, timing advance (TADV), and/or AoA

[0192] Here, TADV may be divided into Type 1 and Type 2 as follows.

$$T_{ADV}\ \text{Type 1} = (ng\text{-}eNB\ Rx\text{-}Tx\ \text{time difference}) + (UE\ E\text{-}UTRA\ Rx\text{-}Tx\ \text{time difference})$$

$$TADV\ \text{Type 2} = ng\text{-}eNB\ Rx\text{-}Tx\ \text{time difference}$$

[0193] AoA may be used to measure the direction of the UE. AoA is defined as the estimated angle of the UE counterclockwise from the eNB/TP. In this case, a geographical reference direction may be north. The eNB/TP may use a UL signal such as an SRS and/or a DMRS for AoA measurement. The accuracy of measurement of AoA increases as the arrangement of an antenna array increases. When antenna arrays are arranged at the same interval, signals received at adjacent antenna elements may have constant phase rotate.

Multi RTT (Multi-Cell RTT)

[0194] FIG. **11** is a diagram illustrating an exemplary multi-round trip time (multi-RTT) positioning method to which the embodiment is applicable.

[0195] Referring to FIG. **11**(*a*), an exemplary RTT procedure is illustrated, in which an initiating device and a responding device perform ToA measurements, and the responding device provides ToA measurements to the initiating device, for RTT measurement (calculation). The initiating device may be a TRP and/or a UE, and the responding device may be a UE and/or a TRP.

[0196] In operation **1301** according to the embodiment, the initiating device may transmit an RTT measurement request, and the responding device may receive the RTT measurement request.

[0197] In operation **1303** according to the embodiment, the initiating device may transmit an RTT measurement signal at t**0** and the responding device may acquire a ToA measurement t**1**.

[0198] In operation **1305** according to the embodiment, the responding device may transmit an RTT measurement signal at t**2** and the initiating device may acquire a ToA measurement t**3**.

[0199] In operation **1307** according to the embodiment, the responding device may transmit information about [t**2**-t**1**], and the initiating device may receive the information and calculate an RTT by Equation 6. The information may be transmitted and received based on a separate signal or in the RTT measurement signal of operation **1305**.

$$RTT = t_3 - t_0 - [t_2 - t_1] \quad \text{[Equation 6]}$$

[0200] Referring to FIG. **11**(*b*), an RTT may correspond to a double-range measurement between two devices. Positioning estimation may be performed from the corresponding information, and multilateration may be used for the positioning estimation. d_1, d**2**, and d**3** may be determined based on the measured RTT, and the location of a target device may be determined to be the intersection of the circumferences of circles with radiuses of d_1, d**2**, and d**3**, in which BS**1**, BS**2**, and BS**3** (or TRPs) are centered, respectively.

2.7. Sounding Procedure

[0201] In a wireless communication system to which the embodiment is applicable, an SRS for positioning may be used.

[0202] An SRS-Config information element (IE) may be used to configure SRS transmission. (A list of) SRS resources and/or (a list of) SRS resource sets may be defined, and each resource set may be defined as a set of SRS resources.

[0203] The SRS-Config IE may include configuration information on an SRS (for other purposes) and configuration information on an SRS for positioning separately. For example, configuration information on an SRS resource set for the SRS (for other purposes) (e.g., SRS-ResourceSet) and configuration information on an SRS resource set for the SRS for positioning (e.g., SRS-PosResourceSet) may be included separately. In addition, configuration information on an SRS resource for the SRS (for other purposes) (e.g., SRS-ResourceSet) and configuration information on an SRS resource for the SRS for positioning (e.g., SRS-PosResource) may be included separately.

[0204] An SRS resource set for positioning may include one or more SRS resources for positioning. Configuration information on the SRS resource set for positioning may include: information on an identifier (ID) that is assigned/allocated/related to the SRS resource set for positioning; and information on an ID that is assigned/allocated/related to each of the one or more SRS resources for positioning. For example, configuration information on an SRS resource for positioning may include an ID assigned/allocated/related to a UL resource. In addition, each SRS resource/SRS resource set for positioning may be identified based on each ID assigned/allocated/related thereto.

[0205] The SRS may be configured periodically/semi-persistently/aperiodically.

[0206] An aperiodic SRS may be triggered by DCI. The DCI may include an SRS request field.

[0207] Table 6 shows an exemplary SRS request field.

TABLE 6

Value of SRS request field	Triggered aperiodic SRS resource set(s) for DCI format 0_1, 0_2, 1_1, 1_2, and 2_3 configured with higher layer parameter srs-TPC-PDCCH-Group set to 'typeB'	Triggered aperiodic SRS resource set(s) for DCI format 2_3 configured with higher layer parameter srs-TPC-PDCCH-Group set to 'typeA'
00	No aperiodic SRS resource set triggered	No speriodic SRS resource set triggered
01	SRS resource set(s) configured by SRS-ResourceSet with higher layer parameter aperiodicSRS-ResourceTrigger set to 1 of an entry in aperiodicSRS-ResourceTriggerList set to 1 SRS resource set(s) configured by SRS-PosResourceSet with an entry in aperiodicSRS-ResourceTriggerList set to 1 when triggered by DCI formats	SRS resource set(s) configured with higher layer parameter usage in SRS-ResourceSet set to 'antennaSwitching' and resourceType in SRS-ResourceSet set to 'aperiodic' for a 1st set of serving cells configured bry higher layers

TABLE 6-continued

Value of SRS request field	Triggered aperiodic SRS resource set(s) for DCI format 0_1, 0_2, 1_1, 1_2, and 2_3 configured with higher layer parameter srs-TPC-PDCCH-Group set to 'typeB'	Triggered aperiodic SRS resource set(s) for DCI format 2_3 configured with higher layer parameter srs-TPC-PDCCH-Group set to 'typeA'
	0_1, 0_2, 1_1, and 1_2	
10	SRS resource set(s) configured by SRS-ResourceSet with higher layer parameter aperiodicSRS-ResourceTrigger set to 2 or an entry in aperiodicSRS-ResourceTriggerList set to 2 SRS resource set(s) configured by SRS-PosResourceSet with an entry in aperiodicSRS-ResourceTriggerList set to 2 when triggered by DCI formats 0_1, 0_2, 1_1, and 1_2	SRS resource set(s) configured with higher layer parameter usage in SRS-ResourceSet set to 'antennaSwatching' and resourceType in SRS-ResourceSet set to 'aperiodic' for a 2nd set of serving cells configured by higher layers
11	SRS resource set(s) configured by SRS-ResourceSet with higher layer parameter aperiodicSRS-ResourceTrigger set to 3 or an entry in aperiodicSRS-ResourceTriggerList set to 3 SRS resource set(s) configured by SRS-FosResourceSet with an entry in aperiodicSRS-ResourceTriggerList set to 3 when triggered by DCI formats 0_1, 0_2, 1_1, and 1_2	SRS resource set(s) configured with higher layer parameter usage in SRS-ResourceSet set to 'antennaSwitching' and resourceType in SRS-ResourceSet set to 'aperiodic' for a 3rd set of serving cells configured by higher layers

[0208] In Table 6 srs-TPC-PDCCH-Group is a parameter for setting the triggering type for SRS transmission to type A or type B, aperiodicSRS-ResourceTriggerList is a parameter for configuring an additional list of DCI code points where the UE needs to transmit the SRS according to the SRS resource set configuration, aperiodicSRS-Resource-Trigger is a parameter for configuring a DCI code point where the SRS needs to be transmitted according to the SRS resource set configuration, and resourceType is a parameter for configuring (periodic/semi-static/aperiodic) time domain behavior of the SRS resource configuration.

3. Embodiment

[0209] A detailed description will be given of the embodiment based on the above technical ideas. The afore-described contents of Section 1 and Section 2 are applicable to the embodiment described below. For example, operations, functions, terminologies, and so on which are not defined in the embodiment may be performed and described based on Section 1 and Section 2.

[0210] Symbols/abbreviations/terms used in the description of the embodiment may be defined as follows.

[0211] A/B/C: A and/or B and/or C

[0212] LMF: location management function

[0213] PRS: positioning reference signal

[0214] SRS: sounding reference signal. According to the embodiment, the SRS may be used for UL channel estimation based on multi-input multi-output (MIMO) and positioning measurement. In other words, according to the embodiment, the SRS may include a normal SRS and a positioning SRS. According to the embodiment, the positioning SRS may be understood as a UL RS configured and/or used for UE positioning. According to the embodiment, the normal SRS is different from the positioning SRS. Specifically, the normal SRS may be understood as a UL RS configured and/or used for UL channel estimation (additionally or alternatively, the normal SRS may be understood as a UL RS configured and/or used for UL channel estimation and positioning). According to the embodiment, the positioning SRS may also be referred to as an SRS for positioning. In the description of the embodiment, the following terms: 'positioning SRS' and 'SRS for positioning' may be used interchangeably and understood to have the same meaning. According to the embodiment, the normal SRS may also be referred to as a legacy SRS, a MIMO SRS, an SRS for MIMO, or the like. In the description of the embodiment, the following terms: 'normal SRS', 'legacy SRS', 'MIMO SRS', and 'SRS for MIMO' may be used interchangeably and understood to have the same meaning. For example, the normal SRS and the positioning SRS may be separately configured/indicated. For example, the normal SRS and the positioning SRS may be configured/indicated by different IEs (information elements) of higher layers. For example, the normal SRS may be configured based on SRS-resource, and the positioning SRS may be configured based on SRS-PosResource.

[0215] TRP: transmission and reception point (TP: transmission point)

[0216] In the description of the embodiment, a BS may be understood as a comprehensive term including a remote radio head (RRH), an eNB, a gNB, a TP, a reception point (RP), a relay, and the like.

[0217] In the description of the embodiment, the expression 'greater than/above A' may be replaced with the expression 'above/greater than A'.

[0218] In the description of the embodiment, the expression 'less than/below B' may be replaced with the expression 'below/less than B'.

[0219] The methods proposed in the embodiment described below may be applied to transmission of a PRS and/or an SRS for positioning.

[0220] The methods proposed in the embodiment described below may be applied regardless of RRC states.

[0221] According to the embodiment, the BS and/or UE may request a preferred value for a configuration parameter related to the methods proposed in the embodiment described below. According to the embodiment, the UE and/or BS may configure/transmit the configuration parameter in consideration of the requested preferred value. For example, the configuration parameter may be configured based on the requested preferred value, but the present disclosure is not limited thereto.

[0222] 3GPP has standardized and developed technologies to support various devices such as machine type communication (MTC) devices and narrowband Internet of Things (NB-IoT) devices as well as existing portable UE devices. For example, reduced capability (RedCap) NR, which allows gains in terms of cost by lowering the capability of current NR, is less sensitive to data rates, and requires low latency, has been introduced as one of these technologies in Rel-17.

[0223] UEs that support RedCap may be used for wearables, industrial wirelee sensors, video surveillance, etc. The UE that supports RedCap may decrease the maximum support number for the bandwidth reduction, MIMO layers, modulation orders, etc., reduce the number of Rx branches compared to the current NR UE, and support half-duplex (HD) on all bands, thereby obtaining capability gains.

[0224] The capabilities of the RedCap UE may be summarized as follows.

- [0225] Reduced maximum UE bandwidth: A maximum of 20 MHz for FR1 and a maximum of 100 MHz for FR2
- [0226] Reduced minimum number of UE Rx branches: For example, 1 or 2
- [0227] Reduced maximum number of MIMO layers: For example, 1,2
- [0228] Relaxed maximum modulation order: For example, 64 QAM
- [0229] Half-duplex FDD type A

[0230] When a UE supporting legacy NR is referred to as a normal UE (or legacy UE), the RedCap UE may have performance degradation in terms of (positioning) accuracy because the RedCap UE supports a small bandwidth.

[0231] According to the embodiment, there are provided a method for band swithcing/hopping to obtain a frequency diversity gain in an SRS for positioning (SRSp) for a RedCap UE and a method of transmitting a burst SRSp to obtain a combing gain in the time domain.

[0232] According to one embodiment, a BW for the RedCap UE may be configured within one PFL, so that legacy UEs (PFL) and RedCap UEs (BW) may be supported simultaneously for positioning by a base station (BS)/location management function (LMF).

[0233] FIG. **12** is a simplified diagram illustrating an operating method of a UE, a TRP, a location server, and/or an LMF according to the embodiment.

[0234] Referring to FIG. **12**, in operation **1301** according to the embodiment, the location server and/or the LMF may transmit configuration indicated to the UE and the UE may receive the configuration information.

[0235] In operation **1303** according to the embodiment, the location server and/or the LMF may transmit reference configuration information to the TRP and the TRP may receive the reference configuration information. In operation **1305** according to the embodiment, the TRP may transmit the reference configuration information to the UE and the UE may receive the reference configuration information. In this case, operation **1201** according to the embodiment may be omitted.

[0236] In contrast, operations **1303** and **1305** according to the embodiment may be omitted. In this case, operation **1301** according to the embodiment may be performed.

[0237] That is, operation **1301** according to the embodiment, and operations **1303** and **1205** according to the embodiment may be selectively performed.

[0238] In operation **1307** according to the embodiment, the TRP may transmit a signal related to the configuration information and the UE may receive the signal related to the configuration information. For example, the signal related to the configuration information may be a signal for positioning of the UE.

[0239] In operation **1309** according to the embodiment, the UE may transmit a signal related to positioning to the TRP and the TRP may receive the signal related to positioning. In operation **1411** according to the embodiment, the TRP may transmit the signal related to positioning to the location server and/or the LMF and the location server and/or the LMF may receive the signal related to positioning.

[0240] In operation according to the embodiment, the UE may transmit the signal related to positioning to the location server and/or the LMF and the location server and/or the LMF may receive the signal related to positioning. In this case, operations **1309** and **1311** according to the embodiment may be omitted.

[0241] In contrast, operation **1313** according to the embodiment may be omitted. In this case, operations **1309** and **1311** according to the embodiment may be performed.

[0242] That is, operations **1309** and **1311** according to the embodiment, and operation **1213** according to the embodiment may be selectively performed.

[0243] According to the embodiment, the signal related to positioning may be obtained based on the configuration information and/or the signal related to the configuration information.

[0244] FIG. **13** is a simplified diagram illustrating an operating method of a UE, a TRP, a location server, and/or an LMF according to the embodiment.

[0245] Referring to FIG. **13**(*a*), in operation **1401**(*a*) according to the embodiment, the UE may receive configuration information.

[0246] In operation **1403**(*a*) according to the embodiment, the UE may receive a signal related to the configuration information.

[0247] In operation **1405**(*a*) according to the embodiment, the UE may transmit information related to positioning.

[0248] Referring to FIG. **13**(*b*), in operation **1401**(*b*) according to the embodiment, the TRP may receive configuration information from the location server and/or the LMF and transmit the configuration information to the UE.

[0249] In operation **1403**(*b*) according to the embodiment, the TRP may transmit a signal related to the configuration information.

[0250] In operation **1405**(*b*) according to the embodiment, the TRP may receive information related to positioning and transmit the information related to positioning to the location server and/or the LMF.

[0251] Referring to FIG. **13**(*c*), in operation **1401**(*c*) according to the embodiment, the location server and/or the LMF may transmit configuration information.

[0252] In operation **1405**(*c*) according to the embodiment, the location server and/or the LMF may receive information related to positioning.

[0253] For example, the above-described configuration information may be understood as relating to reference configuration (information) or one or more pieces of information that the location server, the LMF, and/or the TRP transmits to/configures for the UE and/or may be understood as the reference configuration (information) or one or more pieces of information that the location server, the LMF, and/or the TRP transmits to/configures for the UE, in a description of the embodiment below.

[0254] For example, the above signal related to positioning may be understood as a signal related to one or more pieces of information that the UE reports and/or a signal including one or more pieces of information that the UE reports, in a description of the embodiment below.

[0255] For example, in a description of the embodiment below, the BS, the gNB, and the cell may be replaced with the TRP, the TP, or any device serving equally as the TRP or the TP.

[0256] For example, in a description of the embodiment below, the location server may be replaced with the LMF and any device serving equally as the LMF.

[0257] More detailed operations, functions, terms, etc. in operation methods according to the embodiment may be performed and described based on the embodiment described later. The operation methods according to the embodiment is exemplary and one or more operations in the above-described operation methods may be omitted according to detailed content of each embodiment.

[0258] Hereinafter, the embodiment will be described in detail. It may be understood by those of ordinary skill in the art that the embodiment described below may be combined in whole or in part to implement other embodiments unless mutually exclusive.

Configuration of Positioning Frequency Layer (PFL)

[0259] For example, the LMF may transmit configuration information (e.g., subcarrier spacing (SCS), bandwidth (BW), start PRB (physical resource block), point A, comb size, cyclic prefix (CP), etc.) about up to four PFLs to the legacy UE. For example, information transmitted by the LMF may include information about PRS resources and/or sets that can be transmitted to TRPs for each PFL configuration.

[0260] According to one embodiment, a method such as the following method to be described below may be considered for the RedCap UE.

[0261] According to one embodiment, a method may be considered in which the RedCap UEs can use PFL configuration information configured for the legacy UEs without change in order to be compatible with overhead of the configuration and compatibility with the legacy UEs.

[0262] According to one embodiment, a method may be considered in which, for the RedCap UE, the LMF can more flexibly allocate separate PFL(s) and TRP/PRS resource set/PRS resource(s) related to the PFL(s).

[0263] In the description of one embodiment, the bandwidth of the PFL configured for the legacy UE may be defined as a common bandwidth (cBW) and/or a normal bandwidth (nBW). In the description of one embodiment, cBW and nBW may be understood as the same.

[0264] In the description of one embodiment, the bandwidth of the PFL configured for the RedCap UE may be defined as 'rBW' (bandwidth for RedCap, reduced bandwidth) and/or 'pBW' (positioning BW). In the description of one embodiment, rBW and pBW may be understood as the same.

Method #1: Configuration of Plurality of Subbands for Wideband Configuration and Hopping

[0265] According to Method #1, a wideband, which is an area where not only a plurality of subbands in which UE's substantial hopping occurs, but also the corresponding frequency hopping/switching can occur.

[0266] According to one embodiment, the configuration of the wideband may be instructed/configured to the UE from the LMF based on one or more pieces of information from among SCS, BW, start PRB, point A, comb size, and CPs.

[0267] According to one embodiment, in order to reduce overhead for configuration, compatibility with legacy UEs and overhead for PRS transmission of base stations, the RedCap UE may consider the PFL configuration instructed/configured for the legacy UE as a wideband.

(Sub Narrow Band Configuration for Internal Frequency Hopping)

[0268] In one embodiment, since the RedCap UE uses a reduced BW, there may be multiple rBWs within the nBW configured for the legacy UE.

[0269] According to one embodiment, the LMF may configure/instruct that the PRS can be transmitted in a specific single rBW by allowing the RedCap UE to use all of the multiple rBWs and/or assigning an ID for each rBW and additionally transmitting the ID to the UE.

[0270] According to one embodiment, the UE may expect that the PRS is transmitted only within the configured/instructed specific rBW.

[0271] The UE, LMF, and/or BS may operate in different ways depending on various methods related to the following embodiments to be described below.

[0272] According to one embodiment, the LMF may transmit information about UE's rBW to a serving cell and/or neighbor cell through the NRPPa message for UE's PRS reception expectation and a UE's smooth positioning procedure.

[0273] According to one embodiment, the information about the rBW may include a start PRB for a plurality of pBWs and/or a size for each pBW according to one embodiment to be described below. Additional information according to one embodiment to be described below may also be included.

[0274] FIG. **14** is a diagram illustrating an example of a method of configuring the bandwidth (BW) according to one embodiment.

[0275] FIG. **14** is a diagram illustrating an example of a method for configuring the rBW according to one embodiment.

[0276] Referring to FIG. **14**, according to one embodiment, there may be four methods for configuring the rBW.

[0277] According to one embodiment, in all four methods, a plurality of rBWs may be configured to be included in a single PFL (e.g., PFL #**0**, cBW) within a frequency domain.

[0278] According to one embodiment, one PFL may be configured based on an offset (e.g., a PRB offset) configured based on Point A.

[0279] According to one embodiment, in all four methods, since multiple rBWs are included in one PFL in the frequency domain, the offset for the PFL can be understood as a frequency offset that is commonly applied to all four methods.

[0280] The four methods according to one embodiment can be as follows.

FIG. **14**(A)—Method 1

[0281] According to one embodiment, without a separate PRB offset, the LMF can configure/instruct a single size value to perform allocation for enabling each rBW to have the same size based on the lowest PRB of the cBW. According to one embodiment, multiple rBWs can all have the same size and can be configured to start from the lowest PRB of the cBW.

FIG. **14**(B)—Method 2

[0282] According to one embodiment, only a common PRB offset value (an offset indicated (or instructed) based on the lowest PRB of the cBW) and a single BW size value are configured/indicated from the cBW, so that rBWs may be configured consecutively in the frequency domain.

FIG. **14**(C)—Method 3

[0283] According to one embodiment, the sizes of rBWs may be individually configured/indicated for each rBW, and a single PRB offset may be configured/indicated to the UE.

FIG. **14**(C)—Method 4

[0284] According to one embodiment, the PRB offset and the BW size may be configured/indicated for each rBW to implement a flexible configuration of each rBW.

[0285] According to one embodiment, the plurality of rBWs to be configured may be distinguished by an ID (for example, rBW #**0**, rBW #**1**, . . . , rBW #M (or rBW #N, rBW #O, rBW #P)).

[0286] According to one embodiment, as shown in Option #**2** to be described below, the ID may allow the LMF to configure/indicate a specific rBW within a plurality of rBWs.

[0287] According to one embodiment, the UE may be used to perform hopping/switching of the rBW by referring to the ID for each rBW.

[0288] According to one embodiment, the UE may report the maximum number of supported pBWs supported within the cBW when reporting the capability, and the LMF may indicate/configure the pBWs according to the corresponding capability. According to one embodiment, the corresponding information may also be delivered to all TRPs present in the assistance information.

[0289] According to one embodiment, the PRB offset and/or BW size configured/indicated for each PFL may be individually configured per PFL or may be commonly applied to all PFLs. According to one embodiment, this may be based on a trade-off relationship between a flexible configuration and a configuration overhead.

[0290] According to one embodiment, one of Methods 1 to 4 may be applied to the method for configuring/indicating the rBW for each PFL, and/or one of Methods 1 to 4 may also be applied to all PFLs. According to one embodiment, this may be based on a trade-off relationship between a flexible configuration and a configuration overhead.

[0291] Additionally and/or separately, according to one embodiment, the SCS, comb size, and CP may all be applied equally to multiple rBWs existing in a single PFL in order to configure legacy UEs and unified PRS resources. According to one embodiment, the offset for rBW may be an offset from the CRB of PFL #**0** and/or an offset from Point A.

[0292] According to one embodiment, when the IDs of rBWs are automatically configured based on the lowest PRB and/or the highest PRB according to FIGS. **14**(A) to **14**(D), and/or, when the sizes and offsets of rBWs are individually configured for each rBW as in (D), the IDs may be assigned based on the selection of the LMF.

[0293] According to one embodiment, the number of rBWs existing in a single cBW of the UE capability may be counted in the order of indexing according to FIGS. **14**(A) to **14**(D), and/or the rBWs to be supported by the UE may be different based on the total number of counting for the descending/ascending numerical order of the indices of the rBWs, and the UE may report this maximum value to the LMF for each single PFL and/or for all PFLs (per UE).

Option #1: PRS Transmission in Multiple rBWs

[0294] According to Option #1, regardless of the number of rBWs existing in nBW, the serving cell and the neighbor cells are expected to receive PRSs from all PRS resources within the indicated/configured PFL. The UE may, depending on multiple implementations, monitor only specific rBWs among the multiple rBWs to expect PRS reception, or may, depending on the UE capability, expect to simultaneously receive PRSs from multiple rBWs.

[0295] And/or, according to one embodiment, the UE may perform measurement by performing hopping between rBWs in a preset pattern for the multiple rBWs, and may report the best result, or may report measurement results for all measured rBWs together with the ID of each rBW.

[0296] And/or, according to one embodiment, the UE may report side information (i.e., additional information) for enabling the LMF to recognize the reliability of the reported measurement, may report information as to whether switching/hopping was performed, may report the total number of rBws used for the executed measurement, and/or may report the level of rBWs used for the executed measurement.

Option #2: PRS Transmission within a Single rBW

[0297] According to Option #2, PRS transmission may be performed within a single rBW among multiple rBWs existing within the nBW.

[0298] According to one embodiment, the LMF may basically inform the UE of the rBW in which the PRS is transmitted through an ID for a specific rBW. And/or, according to one embodiment, the LMF may enable the UE to perform switching/hopping of the plurality of rBWs using a preset pattern in order to obtain a diversity gain from a frequency domain through a preset configuration, so that the LMF may configure the UE to perform PRS measurement and report.

[0299] According to one embodiment, the method of configuring/instructing the pattern may operate based on the following method to be described below. For activation/deactivation for FH switching/hopping according to one embodiment to be described below, the UE may expect to monitor a specific rBW according to the pattern without separate activation/deactivation for the configured pattern, and/or the LMF and/or the base station (BS) may separately provide the UE with activation/deactivation for FH through separate signaling. According to an embodiment, triggering/activation/deactivation may be configured through RRC/MAC-CE/DCI (radio resource control/medium access control-control element/DCI) by the LMF designed to transmit such configuration information to the base station (BS). And/or, according to an embodiment, triggering/activation/deactivation may also be performed through an LPP message such as a measurement request.

[0300] According to one embodiment, the rBW switching/hopping to be described below may be performed in units of RBWol(s), in units of R slot(s), in units of R subframe(s), and/or in units of R frame(s) according to the following method according to one embodiment to be described below.

[0301] According to one embodiment, the value of 'R' may be set to the UE from the LMF. According to one embodiment, the value of 'R' may have a value starting from 0 (an integer value greater than or equal to 0), and if the value of 'R' is zero '0', it is considered that FH is not indicated and the UE may not expect rBW switching.

[0302] According to one embodiment, a point where a specific operation (e.g., switching/hopping) starts may be used as a reference point of the start symbol/slot/subframe/frame of the nearest PRS after occurrence of the measurement request. And/or, according to one embodiment, switching/hopping may be performed in units of a single slot, in units of a single subframe, and/or in units of a single frame by a default without a separate duration configuration.

[0303] In one embodiment, rBW switching/hopping may continue until the end of PRS measurement, unless there is a separate deactivation.

[0304] In one embodiment, the configured/indicated rBW switching/hopping pattern to be described below can be commonly applied to each legacy PFL, and/or the pattern may be configured/indicated for each PFL.

[0305] According to one embodiment, in order to support not only rBW switching/hopping within a single PFL but also rBW switching/hopping within multiple PFLs, in addition to pattern indication related information for a method according to one embodiment to be described below, an ID for a legacy PFL (cBW) may be indicated so that not only rBW switching/hopping within a single PFL but also switching between legacy PFLs may be indicated.

[0306] According to one embodiment, switching between PFLs may also be configured/indicated by a separate section. According to one embodiment, a UE may perform switching/hopping to a PFL configured/indicated within the corresponding configuration section. According to one embodiment, the switching/hopping method for each PFL may follow the following method according to one embodiment to be described below.

[0307] According to one embodiment, switching/hopping between the corresponding legacy PFLs may be applied to all methods and/or options described in one embodiment. According to one embodiment, in order to minimize the operation of BWP switching for FH, the position where FH starts within the MG may be determined to be an active BWP (or an rBW including an active BWP).

Alt. #1: Direct Pattern Instruction Using rBW ID

[0308] According to Alt. #1, according to one embodiment, rBW switching/hopping patterns may be instructed/configured to the UE using a combination of rBW IDs.

[0309] According to one embodiment, the LMF may configure/instruct the UE about M consecutive rBW IDs, and the UE may expecte to perform rBW switching/hopping according to the M patterns.

[0310] For example, if the number of rBMs configured in the UE is 4 and 4 rBW ID combination (i.e., {0,2,1,3}) are configured, the UE may continuously perform rBW switching/hopping for rBWs (#0, 2, 1, 3) after lapse of the duration described above (i.e., rBW #2→rBW #1→rBW #3 and/or rBW #0→rBW #2→rBW #1→rBW #3). For example, the UE may follow the corresponding pattern until the measurement is finished.

[0311] Additionally and/or separately, according to one embodiment, specific patterns may be tabulated and predefined. According to one embodiment, the LMF may configure/instruct the UE about the index for the specific pattern described in the table. In one embodiment, the UE may identify the specific pattern based on the table and the index.

[0312] In one embodiment, a flexible configuration in which the LMF freely allows rBW switching/hopping or does not allow such switching/hopping to a specific section can be allowed.

Alt. #2: Indicate the Offset Between rBWs and/or Configure/Indicate the ID for Initial rBW

[0313] According to Alt. #2, only the offset value for switching/hopping between rBWs may be transmitted by the LMF to the UE and/or the ID of a separate initial rBW may be transmitted to the UE.

[0314] In the former case, according to one embodiment, the ID of the initial rBW may be determined by implementation of the UE or may be operated as rBW #0 by default. According to one embodiment, the subsequent switching/hopping may be performed according to the offset configured/indicated for each section according to the above-described embodiment.

[0315] According to one embodiment, the switching/hopping may mean the offset between rBW ID indices from a virtual perspective and/or may be performed in units of an offset for the indices arranged in descending and/or ascending order based on the rBW closest to the lowest/highest PRB of the cBW from a physical perspective.

[0316] For example, if a value for the offset is set to '2' by the LMF as a single value, the UE may perform switching/hopping through +2 and/or −2 modulo operation (modulo operation based on the indicated offset value) for the index of the rBW ID after the duration described above in the initial rBW #0 or the default rBW #n. For example, if the initial UE is present in rBW #0, the UE may perform rBW switching/hopping in the order of rBW #2→rBW #0→rBW #2.

[0317] Additionally and/or separately, according to one embodiment, the LMF may additionally configure/instruct the initial rBW, and/or the BW in which data is transmitted without a separate configuration may be used as the default rBW and/or rBW #0 may be used as the default rBW.

[0318] According to one embodiment, the UE may not expect PRS reception for PRS resources that overlap with the switching time of the rBW switching/hopping.

Alt. #3: Predefined Hopping Pattern

[0319] According to Alt. #3, a hopping pattern may be determined in advance. According to one embodiment, the Alt. #3 operation may depend on the total number of rBWs in which switching/hopping of the rBW of the UE may occur. According to one embodiment, a hopping pattern may be predefined for each number of rBWs supported by the UE.

[0320] For example, if there are 4 rBWs per cRB and the UE actually performs switching/hopping for 3 rBWs based on the UE capability, the UE may implicitly use a hopping pattern corresponding to 3. For example, if the number of rBWs where hopping occurs is 2, 3, or 4, and the patterns are predefined as {0,1}, {0,2,1}, {0,2,1,3} for each pattern, then the UE may perform rBW switching/hopping in the order of rBW #0→rBW #2→rBW #1. For example, the corresponding value may be regarded as an offset for default rBW #n.

[0321] In one embodiment, default rBW #n may be the ID of the rBW selected by the UE for initial PRS monitoring.

[0322] For example, when the UE performs switching/hopping for 4 rBWs, the UE may perform such switching/hopping in the order of rBW #n→rBW #(mod (n+2,4)) →rBW #(mod(n+1,4))→rBW #(mod (n+3,4)).

Method #2: Separated Configuration of PFL for RedCap UE (Separate Configuration for SCS, BW. Start PRB, Point A, Comb Size, CP)

[0323] According to Method #2, separate PFL(s) may be configured/instructed by the LMF for RedCap UE.

[0324] According to one embodiment, a flexible configuration may be instructed by the LMF, and simultaneous transmission of PRS in different time/frequency resources may be supported for legacy UEs and RedCap UEs by the base station (BS).

[0325] In one embodiment, "#of PFLs for RedCap UE (the number of PFLs for RedCap UE)" may vary depending on the UE capability. In one embodiment, the UE may transmit the maximum number of rPFL(s) (e.g., 'RedCap PFL', reduced PFL) and/or related information supported when the UE capability is reported.

[0326] According to one embodiment, the LMF may configure separate rPFL(s) for the RedCap UE based on the reported capability. In one embodiment, the configuration of the legacy PFL may be used as a parameter required for such configuration, but each value may be configured/instructed to the UE by the LMF as a different value for each rPFL(s).

[0327] According to one embodiment, the capability report to the LMF may be performed regarding whether the UE supports separate PFL configuration. In one embodiment, the UE may need to report the maximum number of PFLs supported. And/or, according to one embodiment, information as to whether the band for multiple FH hopping actions is supported within a single PFL and/or information about the maximum number of supported PFLs may also be reported.

[0328] According to one embodiment, each rPFL may be configured to be smaller than the BW size supported by the RedCap UE.

[0329] According to one embodiment, the base station (BS) may assign an ID to each rPFL. In one embodiment, the method of assigning the corresponding ID may be based on the method for assigning the rBW's ID described in Method #**1**.

[0330] According to one embodiment, in the case of the UE capable of monitoring multiple rPFLs at the same time, the UE capability may be reported to the LMF. According to one embodiment, the LMF may be requested from the base station (BS) through the corresponding information as to whether to simultaneously transmit multiple rPFLs.

[0331] According to one embodiment, if the UE is capable of monitoring multiple rPFLs, the LMF and/or the base station (BS) may configure/instruct whether to simultaneously transmit multiple rPFLs. According to one embodiment, if simultaneous transmission in multiple rPFLs is instructed, the base station (BS) may transmit PRSs through multiple rPFLs, and the UE may monitor/receive PRSs based on the action of monitoring multiple rPFLs.

[0332] According to one embodiment, if transmission is instructed within a specific single rPFL among multiple rPFLs, the base station (BS) may transmit PRSs in a specific single rPFL, and the UE may monitor/receive PRSs based on the operation of monitoring a specific single rPFL. For example, information about the specific single rPFL may be transmitted to the UE, and the UE may perform monitoring for the specific single rPFL based on the transmitted information.

[0333] According to one embodiment, for frequency diversity, the switching/hopping method described in the rPFLs and/or rBWs in Method #**1** can be commonly applied to such switching/hopping between rPFs described in Method #**2**.

[0334] According to one embodiment, all base stations associated with PRS resources existing in the pattern may perform PRS transmission from single and/or multiple rPFLs according to the configured pattern, and/or may perform PRS transmission from all PFLs according to the configured pattern.

[0335] According to one embodiment, the UE may perform rPFL switching/hopping and expect PRS measurement according to the configured rPFL switching/hopping pattern. According to one embodiment, the UE may not expect PRS reception for the switching/hopping time required for rPFL switching/hopping and the overlapped PRS resources.

[0336] According to one embodiment, the detailed rPFL switching/hopping configuration and the ID configuration may be directly applied to the ID mapping and pattern indication method for nBW described in Method #**1**.

[0337] FIG. **15** is a diagram briefly illustrating a method for operating the UE and the network nodes according to one embodiment.

[0338] FIG. **16** is a flowchart illustrating a method for operating the UE according to one embodiment.

[0339] FIG. **17** is a flowchart illustrating a method for operating the network node according to one embodiment. For example, the network node may be a TP, a base station (BS), a cell, a location server, an LMF and/or any device that performs the same task.

[0340] Referring to FIGS. **15** to **17**, in operations **1501**, **1601**, and **1701** according to one embodiment, the network node may transmit assistance data including information related to a configuration for an RS for positioning, and the UE may receive the assistance data.

[0341] According to an embodiment, the information related to the RS configuration may include information related to one or more PFLs.

[0342] In operations **1502**, **1603**, and **1703** according to one embodiment, the UE and the network node may perform communication with the RS for positioning based on assistance data. For example, the RS may include one or more of PRS and SRS (sounding reference signal) for the positioning.

[0343] According to one embodiment, based on information indicating that the UE serves as a RedCap UE, the information related to the RS configuration may include information related to a plurality of first BWs for the RS. According to one embodiment, the plurality of first BWs may be included in one first PFL among PFLs within a frequency domain.

[0344] According to one embodiment, based on information indicating that the UE serves as the RedCap UE, the RS may be communicated in one first specific BW among the plurality of first BWs at one first time instance.

[0345] More specific operations of the UE and/or the network node according to various embodiments of the present disclosure can be described and performed based on the contents of the first to third paragraphs described above.

[0346] It will be appreciated that any of examples of the above-described proposed methods may also be included as one of the embodiments and thus considered to be a kind of proposed method. Further, the above-described proposed methods may be implemented independently, and some of them may also be implemented as a combination (or merged). It may be regulated that the BS indicates to the UE whether to apply the proposed methods (or information about rules for the proposed methods) through a predefined signal (e.g., a physical-layer signal or a higher-layer signal).

4. Exemplary Configurations of Devices Implementing Embodiment of the Disclosure

4.1. Exemplary Configurations of Devices to which Embodiment of the Disclosure is Applied

[0347] FIG. **18** is a diagram illustrating devices that implement various embodiments of the disclosure.

[0348] The devices illustrated in FIG. **18** may be a UE and/or a BS (e.g., an eNB or gNB, or a TP) adapted to perform the above-descried mechanism, or any devices that perform the same operations.

[0349] Referring to FIG. **18**, the device may include a digital signal processor (DSP)/microprocessor **210** and a radio frequency (RF) module (transceiver) **235**. The DSP/microprocessor **210** is electrically coupled to the transceiver **235** and controls the transceiver **235**. The device may further include a power management module **205**, a battery **255**, a display **215**, a keypad **220**, a SIM card **225**, a memory device **230**, an antenna **240**, a speaker **245**, and an input device **250**, depending on a designer's selection.

[0350] Particularly, FIG. **18** may illustrate a UE including a receiver **235** configured to receive a request message from a network and a transmitter **235** configured to transmit timing transmission/reception timing information to the network. These receiver and transmitter may form the transceiver **235**. The UE may further include a processor **210** coupled to the transceiver **235**.

[0351] Further, FIG. **18** may illustrate a network device including a transmitter **235** configured to transmit a request message to a UE and a receiver **235** configured to receive timing transmission/reception timing information from the UE. These transmitter and receiver may form the transceiver **235**. The network may further include the processor **210** coupled to the transceiver **235**. The processor **210** may calculate latency based on the transmission/reception timing information.

[0352] A processor of a UE (or a communication device included in the UE) and/or a BS (or a communication device included in the BS) and/or a location server (or a communication device included in the location server) may operate by controlling a memory, as follows.

[0353] According to the embodiment, the UE or the BS or the location server may include at least one transceiver, at least one memory, and at least one processor coupled to the at least one transceiver and the at least one memory. The at least one memory may store instructions which cause the at least one processor to perform the following operations.

[0354] The communication device included in the UE or the BS or the location server may be configured to include the at least one processor and the at least one memory. The communication device may be configured to include the at least one transceiver or to be coupled to the at least one transceiver without including the at least one transceiver.

[0355] The TP and/or the BS and/or the cell and/or the location server and/or the LMF and/or any device performing the same operation may be referred to as a network node.

[0356] According to one embodiment, one or more processors included in the UE (or one or more processors of the communication device included in the UE) can be configured to receive assistance data including information related to the RS configuration for positioning.

[0357] According to one embodiment, the information related to the RS configuration may include information related to one or more PFLs.

[0358] According to one embodiment, one or more processors included in the UE can be configured to perform communication for the RS for positioning based on the assistance data.

[0359] According to one embodiment, based on information indicating that the UE serves as the RedCap UE, (i) the information related to the configuration for the RS may further include information related to a plurality of first BWs for the RS; and (ii) the RS may be communicated in one first specific BW among the plurality of first BWs at one first time instance, and the plurality of first BWs may be included in a first PFL of the one or more PFLs within the frequency domain.

[0360] According to one embodiment, one or more processors included in a network node (or one or more processors of a communication device included in the network node) may be configured to transmit assistance data including information related to the RS configuration for positioning;

[0361] According to one embodiment, the information related to the RS configuration may include information related to one or more PFLs.

[0362] According to one embodiment, one or more processors included in the network node may be configured to perform communication for the RS for positioning related to the assistance data with the UE.

[0363] According to one embodiment, based on information indicating that the UE serves as the RedCap UE, (i) the information related to the configuration for the RS may further include information related to a plurality of first BWs for the RS; and (ii) the RS may be communicated in one first specific BW among the plurality of first BWs at one first time instance, and the plurality of first BWs may be included in a first PFL of the one or more PFLs within the frequency domain.

[0364] More specific operations of the processor included in the UE and/or the network node according to the above-described embodiment may be described and performed based on the contents of the above-described sections 1 to 3.

[0365] Embodiments may be implemented in combination/conjunction with each other unless they contradict each other. For example, a UE and/or a network node (a processor included in the UE and/or the network node) according to an embodiment may perform combined/merged operations of the embodiments of the above-described sections 1 to 3, unless they contradict each other.

4.2. Example of Communication System to Embodiment of the Disclosure is Applied

[0366] In the present specification, various embodiments of the disclosure have been mainly described in relation to data transmission and reception between a BS and a UE in a wireless communication system. However, various embodiments of the disclosure are not limited thereto. For example, various embodiments of the disclosure may also relate to the following technical configurations.

[0367] The various descriptions, functions, procedures, proposals, methods, and/or operational flowcharts of the various embodiments of the disclosure described in this document may be applied to, without being limited to, a variety of fields requiring wireless communication/connection (e.g., 5G) between devices.

[0368] Hereinafter, a description will be given in more detail with reference to the drawings. In the following drawings/description, the same reference symbols may denote the same or corresponding hardware blocks, software blocks, or functional blocks unless described otherwise.

[0369] FIG. **19** illustrates an exemplary communication system to which various embodiments of the disclosure are applied.

[0370] Referring to FIG. **19**, a communication system **1** applied to the various embodiments of the disclosure includes wireless devices, Base Stations (BSs), and a network. Herein, the wireless devices represent devices performing communication using Radio Access Technology (RAT) (e.g., 5G New RAT (NR)) or Long-Term Evolution (LTE)) and may be referred to as communication/radio/5G devices. The wireless devices may include, without being limited to, a robot **100***a*, vehicles **100***b*-**1** and **100***b*-**2**, an extended Reality (XR) device **100***c*, a hand-held device **100***d*, a home appliance **100***e*, an Internet of Things (IoT) device **100***f*, and an Artificial Intelligence (AI) device/server **400**. For example, the vehicles may include a vehicle having a wireless communication function, an autonomous driving vehicle, and a vehicle capable of performing communication between vehicles. Herein, the vehicles may include an Unmanned Aerial Vehicle (UAV) (e.g., a drone). The XR device may include an Augmented Reality (AR)/Virtual Reality (VR)/Mixed Reality (MR) device and may be implemented in the form of a Head-Mounted Device (HMD), a Head-Up Display (HUD) mounted in a vehicle, a television, a smartphone, a computer, a wearable device, a home appliance device, a digital signage, a vehicle, a robot, etc. The hand-held device may include a smartphone, a smartpad, a wearable device (e.g., a smartwatch or a smartglasses), and a computer (e.g., a notebook). The home appliance may include a TV, a refrigerator, and a washing machine. The IoT device may include a sensor and a smartmeter. For example, the BSs and the network may be implemented as wireless devices and a specific wireless device **200***a* may operate as a BS/network node with respect to other wireless devices.

[0371] The wireless devices **100***a* to **100***f* may be connected to the network **300** via the BSs **200**. An AI technology may be applied to the wireless devices **100***a* to **100***f* and the wireless devices **100***a* to **100***f* may be connected to the AI server **400** via the network **300**. The network **300** may be configured using a 3G network, a 4G (e.g., LTE) network, or a 5G (e.g., NR) network. Although the wireless devices **100***a* to **100***f* may communicate with each other through the BSs **200**/network **300**, the wireless devices **100***a* to **100***f* may perform direct communication (e.g., sidelink communication) with each other without passing through the BSs/network. For example, the vehicles **100***b*-**1** and **100***b*-**2** may perform direct communication (e.g. Vehicle-to-Vehicle (V2V)/Vehicle-to-everything (V2X) communication). The IoT device (e.g., a sensor) may perform direct communication with other IoT devices (e.g., sensors) or other wireless devices **100***a* to **100***f*.

[0372] Wireless communication/connections **150***a*, **150***b*, or **150***c* may be established between the wireless devices **100***a* to **100***f*/BS **200**, or BS **200**/BS **200**. Herein, the wireless communication/connections may be established through various RATs (e.g., 5G NR) such as uplink/downlink communication **150***a*, sidelink communication **150***b* (or, D2D communication), or inter BS communication (e.g. relay, Integrated Access Backhaul (IAB)). The wireless devices and the BSs/the wireless devices may transmit/receive radio signals to/from each other through the wireless communication/connections **150***a* and **150***b*. For example, the wireless communication/connections **150***a* and **150***b* may transmit/receive signals through various physical channels. To this end, at least a part of various configuration information configuring processes, various signal processing processes (e.g., channel encoding/decoding, modulation/demodulation, and resource mapping/demapping), and resource allocating processes, for transmitting/receiving radio signals, may be performed based on the various proposals of the various embodiments of the disclosure.

Example of Wireless Devices to which Embodiment of the Disclosure is Applied

[0373] FIG. **20** illustrates exemplary wireless devices to which various embodiments of the disclosure are applicable.

[0374] Referring to FIG. **20**, a first wireless device **100** and a second wireless device **200** may transmit radio signals through a variety of RATs (e.g., LTE and NR). Herein, {the first wireless device **100** and the second wireless device **200**} may correspond to {the wireless device **100***x* and the BS **200**} and/or {the wireless device **100***x* and the wireless device **100***x*} of FIG. **19**.

[0375] The first wireless device **100** may include one or more processors **102** and one or more memories **104** and additionally further include one or more transceivers **106** and/or one or more antennas **108**. The processor(s) **102** may control the memory(s) **104** and/or the transceiver(s) **106** and may be configured to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. For example, the processor(s) **102** may process information within the memory(s) **104** to generate first information/signals and then transmit radio signals including the first information/signals through the transceiver(s) **106**. The processor(s) **102** may receive radio signals including second information/signals through the transceiver **106** and then store information obtained by processing the second information/signals in the memory(s) **104**. The memory(s) **104** may be connected to the processor (s) **102** and may store a variety of information related to operations of the processor(s) **102**. For example, the memory(s) **104** may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) **102** or for performing the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. Herein, the processor(s) **102** and the memory(s) **104** may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) **106** may be connected to the processor(s) **102** and transmit and/or receive radio signals through one or more antennas **108**. Each of the transceiver(s) **106** may include a transmitter and/or a receiver. The transceiver(s) **106** may be interchangeably used with Radio Frequency (RF) unit(s). In the various embodiments of the disclosure, the wireless device may represent a communication modem/circuit/chip.

[0376] The second wireless device **200** may include one or more processors **202** and one or more memories **204** and additionally further include one or more transceivers **206** and/or one or more antennas **208**. The processor(s) **202** may control the memory(s) **204** and/or the transceiver(s) **206** and may be configured to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. For example, the processor(s) **202** may process information within the memory(s) **204** to generate third information/signals and then transmit radio signals including the third information/signals through the transceiver(s) **206**. The processor(s) **202** may receive radio signals including fourth information/signals through the transceiver(s) **106** and then store information obtained by processing the fourth information/signals in the memory (s) **204**. The memory(s) **204** may be connected to the processor(s) **202** and may store a variety of information related to operations of the processor(s) **202**. For example, the memory(s) **204** may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) **202** or for performing the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. Herein, the processor(s) **202** and the memory(s) **204** may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) **206** may be connected to the processor(s) **202** and transmit and/or receive radio signals through one or more antennas **208**. Each of the transceiver(s) **206** may include a transmitter and/or a receiver. The transceiver(s) **206** may be interchangeably used with RF unit(s). In the various embodiments of the disclosure, the wireless device may represent a communication modem/circuit/chip.

[0377] Hereinafter, hardware elements of the wireless devices **100** and **200** will be described more specifically. One or more protocol layers may be implemented by, without being limited to, one or more processors **102** and **202**. For example, the one or more processors **102** and **202** may implement one or more layers (e.g., functional layers such as PHY, MAC, RLC, PDCP, RRC, and SDAP). The one or more processors **102** and **202** may generate one or more Protocol Data Units (PDUs) and/or one or more Service Data Unit (SDUs) according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. The one or more processors **102** and **202** may generate messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. The one or more processors **102** and **202** may generate signals (e.g., baseband signals) including PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document and provide the generated signals to the one or more transceivers **106** and **206**. The one or more processors **102** and **202** may receive the signals (e.g., baseband signals) from the one or more transceivers **106** and **206** and acquire the PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document.

[0378] The one or more processors **102** and **202** may be referred to as controllers, microcontrollers, microprocessors, or microcomputers. The one or more processors **102** and **202** may be implemented by hardware, firmware, software, or a combination thereof. As an example, one or more Application Specific Integrated Circuits (ASICs), one or more Digital Signal Processors (DSPs), one or more Digital Signal Processing Devices (DSPDs), one or more Programmable Logic Devices (PLDs), or one or more Field Programmable Gate Arrays (FPGAs) may be included in the one or more processors **102** and **202**. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be implemented using firmware or software and the firmware or software may be configured to include the modules, procedures, or functions. Firmware or software configured to perform the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be included in the one or more processors **102** and **202** or stored in the one or more memories **104** and **204** so as to be driven by the one or more processors **102** and **202**. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be implemented using firmware or software in the form of code, commands, and/or a set of commands.

[0379] The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** and store various types of data, signals, messages, information, programs, code, instructions, and/or commands. The one or more memories **104** and **204** may be configured by Read-Only Memories (ROMs), Random Access Memories (RAMs), Electrically Erasable Programmable Read-Only Memories (EPROMs), flash memories, hard drives, registers, cash memories, computer-readable storage media, and/or combinations thereof. The one or more memories **104** and **204** may be located at the interior and/or exterior of the one or more processors **102** and **202**. The one or more memories

104 and **204** may be connected to the one or more processors **102** and **202** through various technologies such as wired or wireless connection.

[0380] The one or more transceivers **106** and **206** may transmit user data, control information, and/or radio signals/channels, mentioned in the methods and/or operational flowcharts of this document, to one or more other devices. The one or more transceivers **106** and **206** may receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document, from one or more other devices. For example, the one or more transceivers **106** and **206** may be connected to the one or more processors **102** and **202** and transmit and receive radio signals. For example, the one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may transmit user data, control information, or radio signals to one or more other devices. The one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may receive user data, control information, or radio signals from one or more other devices. The one or more transceivers **106** and **206** may be connected to the one or more antennas **108** and **208** and the one or more transceivers **106** and **206** may be configured to transmit and receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document, through the one or more antennas **108** and **208**. In this document, the one or more antennas may be a plurality of physical antennas or a plurality of logical antennas (e.g., antenna ports). The one or more transceivers **106** and **206** may convert received radio signals/channels etc. from RF band signals into baseband signals in order to process received user data, control information, radio signals/channels, etc. using the one or more processors **102** and **202**. The one or more transceivers **106** and **206** may convert the user data, control information, radio signals/channels, etc. processed using the one or more processors **102** and **202** from the base band signals into the RF band signals. To this end, the one or more transceivers **106** and **206** may include (analog) oscillators and/or filters.

[0381] According to various embodiments of the disclosure, one or more memories (e.g., **104** or **204**) may store instructions or programs which, when executed, cause one or more processors operably coupled to the one or more memories to perform operations according to various embodiments or implementations of the disclosure.

[0382] According to various embodiments of the disclosure, a computer-readable storage medium may store one or more instructions or computer programs which, when executed by one or more processors, cause the one or more processors to perform operations according to various embodiments or implementations of the disclosure.

[0383] According to various embodiments of the disclosure, a processing device or apparatus may include one or more processors and one or more computer memories connected to the one or more processors. The one or more computer memories may store instructions or programs which, when executed, cause the one or more processors operably coupled to the one or more memories to perform operations according to various embodiments or implementations of the disclosure.

Example of Using Wireless Devices to which Embodiment of the Disclosure is Applied

[0384] FIG. **21** illustrates other exemplary wireless devices to which various embodiments of the disclosure are applied. The wireless devices may be implemented in various forms according to a use case/service (see FIG. **19**).

[0385] Referring to FIG. **21**, wireless devices **100** and **200** may correspond to the wireless devices **100** and **200** of FIG. **19** and may be configured by various elements, components, units/portions, and/or modules. For example, each of the wireless devices **100** and **200** may include a communication unit **110**, a control unit **120**, a memory unit **130**, and additional components **140**. The communication unit may include a communication circuit **112** and transceiver(s) **114**. For example, the communication circuit **112** may include the one or more processors **102** and **202** and/or the one or more memories **104** and **204** of FIG. **19**. For example, the transceiver(s) **114** may include the one or more transceivers **106** and **206** and/or the one or more antennas **108** and **208** of FIG. **19**. The control unit **120** is electrically connected to the communication unit **110**, the memory **130**, and the additional components **140** and controls overall operation of the wireless devices. For example, the control unit **120** may control an electric/mechanical operation of the wireless device based on programs/code/commands/information stored in the memory unit **130**. The control unit **120** may transmit the information stored in the memory unit **130** to the exterior (e.g., other communication devices) via the communication unit **110** through a wireless/wired interface or store, in the memory unit **130**, information received through the wireless/wired interface from the exterior (e.g., other communication devices) via the communication unit **110**.

[0386] The additional components **140** may be variously configured according to types of wireless devices. For example, the additional components **140** may include at least one of a power unit/battery, input/output (I/O) unit, a driving unit, and a computing unit. The wireless device may be implemented in the form of, without being limited to, the robot (**100***a* of FIG. **W1**), the vehicles (**100***b***-1** and **100***b***-2** of FIG. **W1**), the XR device (**100***c* of FIG. **W1**), the hand-held device (**100***d* of FIG. **W1**), the home appliance (**100***e* of FIG. **W1**), the IoT device (**100***f* of FIG. **W1**), a digital broadcast terminal, a hologram device, a public safety device, an MTC device, a medicine device, a fintech device (or a finance device), a security device, a climate/environment device, the AI server/device (**400** of FIG. **W1**), the BSs (**200** of FIG. **W1**), a network node, etc. The wireless device may be used in a mobile or fixed place according to a use-example/service.

[0387] In FIG. **21**, the entirety of the various elements, components, units/portions, and/or modules in the wireless devices **100** and **200** may be connected to each other through a wired interface or at least a part thereof may be wirelessly connected through the communication unit **110**. For example, in each of the wireless devices **100** and **200**, the control unit **120** and the communication unit **110** may be connected by wire and the control unit **120** and first units (e.g., **130** and **140**) may be wirelessly connected through the communication unit **110**. Each element, component, unit/portion, and/or module within the wireless devices **100** and **200** may further include one or more elements. For example, the control unit **120** may be configured by a set of one or more processors. As an example, the control unit **120** may be configured by a set of a communication control processor, an application processor, an Electronic Control Unit (ECU), a graphical processing unit, and a memory control processor. As another example, the memory **130** may be configured by a Random Access Memory (RAM), a Dynamic RAM (DRAM), a Read Only Memory (ROM)), a flash memory, a volatile memory, a non-volatile memory, and/or a combination thereof.

[0388] Hereinafter, an example of implementing FIG. **21** will be described in detail with reference to the drawings.

Example of Portable Device to which Embodiment of the Disclosure is Applied

[0389] FIG. **22** illustrates an exemplary portable device to which various embodiments of the disclosure are applied. The portable device may be any of a smartphone, a smartpad, a wearable device (e.g., a smartwatch or smart glasses), and a portable computer (e.g., a laptop). A portable device may also be referred to as mobile station (MS), user terminal (UT), mobile subscriber station (MSS), subscriber station (SS), advanced mobile station (AMS), or wireless terminal (WT).

[0390] Referring to FIG. **22**, a hand-held device **100** may include an antenna unit **108**, a communication unit **110**, a control unit **120**, a memory unit **130**, a power supply unit **140***a*, an interface unit **140***b*, and an I/O unit **140***c*. The antenna unit **108** may be configured as a part of the communication unit **110**. Blocks **110** to **130**/**140***a* to**140***c* correspond to the blocks **110** to **130**/**140** of FIG. X**3**, respectively.

[0391] The communication unit **110** may transmit and receive signals (e.g., data and control signals) to and from other wireless devices or BSs. The control unit **120** may perform various operations by controlling constituent elements of the hand-held device **100**. The control unit **120** may include an Application Processor (AP). The memory unit **130** may store data/parameters/programs/code/commands needed to drive the hand-held device **100**. The memory unit **130** may store input/output data/information. The power supply unit **140***a* may supply power to the hand-held device **100** and include a wired/wireless charging circuit, a battery, etc. The interface unit **140***b* may support connection of the hand-held device **100** to other external devices. The interface unit **140***b* may include various ports (e.g., an audio I/O port and a video I/O port) for connection with external devices. The I/O unit **140***c* may input or output video information/signals, audio information/signals, data, and/or information input by a user. The I/O unit **140***c* may include a camera, a microphone, a user input unit, a display unit **140***d*, a speaker, and/or a haptic module.

[0392] As an example, in the case of data communication, the I/O unit **140***c* may acquire information/signals (e.g., touch, text, voice, images, or video) input by a user and the acquired information/signals may be stored in the memory unit **130**. The communication unit **110** may convert the information/signals stored in the memory into radio signals and transmit the converted radio signals to other wireless devices directly or to a BS. The communication unit **110** may receive radio signals from other wireless devices or the BS and then restore the received radio signals into original information/signals. The restored information/signals may be stored in the memory unit **130** and may be output as various types (e.g., text, voice, images, video, or haptic) through the I/O unit **140***c*.

Example of Vehicle or Autonomous Driving Vehicle to which Embodiment of the Disclosure is Applied

[0393] FIG. **23** illustrates an exemplary vehicle or autonomous driving vehicle to which various embodiments of the disclosure. The vehicle or autonomous driving vehicle may be implemented as a mobile robot, a car, a train, a manned/unmanned aerial vehicle (AV), a ship, or the like.

[0394] Referring to FIG. **23**, a vehicle or autonomous driving vehicle **100** may include an antenna unit **108**, a communication unit **110**, a control unit **120**, a driving unit **140***a*, a power supply unit **140***b*, a sensor unit **140***c*, and an autonomous driving unit **140***d*. The antenna unit **108** may be configured as a part of the communication unit **110**. The blocks **110**/**130**/**140***a* to **140***d* correspond to the blocks **110**/**130**/**140** of FIG. X**3**, respectively.

[0395] The communication unit **110** may transmit and receive signals (e.g., data and control signals) to and from external devices such as other vehicles, BSs (e.g., gNBs and road side units), and servers. The control unit **120** may perform various operations by controlling elements of the vehicle or the autonomous driving vehicle **100**. The control unit **120** may include an Electronic Control Unit (ECU). The driving unit **140***a* may cause the vehicle or the autonomous driving vehicle **100** to drive on a road. The driving unit **140***a* may include an engine, a motor, a powertrain, a wheel, a brake, a steering device, etc. The power supply unit **140***b* may supply power to the vehicle or the autonomous driving vehicle **100** and include a wired/wireless charging circuit, a battery, etc. The sensor unit **140***c* may acquire a vehicle state, ambient environment information, user information, etc. The sensor unit **140***c* may include an Inertial Measurement Unit (IMU) sensor, a collision sensor, a wheel sensor, a speed sensor, a slope sensor, a weight sensor, a heading sensor, a position module, a vehicle forward/backward sensor, a battery sensor, a fuel sensor, a tire sensor, a steering sensor, a temperature sensor, a humidity sensor, an ultrasonic sensor, an illumination sensor, a pedal position sensor, etc. The autonomous driving unit **140***d* may implement technology for maintaining a lane on which a vehicle is driving, technology for automatically adjusting speed, such as adaptive cruise control, technology for autonomously driving along a determined path, technology for driving by automatically setting a path if a destination is set, and the like.

[0396] For example, the communication unit **110** may receive map data, traffic information data, etc. from an external server. The autonomous driving unit **140***d* may generate an autonomous driving path and a driving plan from the obtained data. The control unit **120** may control the driving unit **140***a* such that the vehicle or the autonomous driving vehicle **100** may move along the autonomous driving path according to the driving plan (e.g., speed/direction control). In the middle of autonomous driving, the communication unit **110** may aperiodically/periodically acquire recent traffic information data from the external server and acquire surrounding traffic information data from neighboring vehicles. In the middle of autonomous driving, the sensor unit **140***c* may obtain a vehicle state and/or surrounding environment information. The autonomous driving unit **140***d* may update the autonomous driving path and the driving plan based on the newly obtained data/information. The communication unit **110** may transfer information about a vehicle position, the autonomous driving path, and/or the driving plan to the external server. The external server may predict traffic information data using AI technology, etc., based on the information collected from vehicles or autonomous driving vehicles and provide the predicted traffic information data to the vehicles or the autonomous driving vehicles.

[0397] In summary, an embodiment may be implemented via a scheduling device and/or UE.

[0398] For example, a device may be a BS, a network node, a transmitting terminal, a receiving terminal, a wireless device, a wireless communication device, a vehicle, a vehicle with autonomous driving capabilities, an unmanned aerial vehicle (UAV), an artificial intelligence (AI) module, a robot, an augmented reality (AR) device, a virtual reality (VR) device, or any other device.

[0399] For example, a UE may be any of a personal digital assistant (PDA), a cellular phone, a personal communication service (PCS) phone, a global system for mobile (GSM) phone, a wideband CDMA (WCDMA) phone, a mobile broadband system (MBS) phone, a smartphone, and a multi mode-multi band (MM-MB) UE.

[0400] A smartphone refers to a terminal taking the advantages of both a mobile communication terminal and a PDA, which is achieved by integrating a data communication function being the function of a PDA, such as scheduling, fax transmission and reception, and Internet connection in a mobile communication UE. Further, an MM-MB terminal refers to a terminal which has a built-in multi-modem chip and thus is operable in all of a portable Internet system and other mobile communication system (e.g., CDMA 2000, WCDMA, and so on).

[0401] Alternatively, the UE may be any of a laptop PC, a hand-held PC, a tablet PC, an ultrabook, a slate PC, a digital broadcasting terminal, a portable multimedia player (PMP), a navigator, and a wearable device such as a smartwatch, smart glasses, and a head mounted display (HMD). For example, a UAV may be an unmanned aerial vehicle that flies under the control of a wireless control signal. For example, an HMD may be a display device worn around the head. For example, the HMD may be used to implement AR or VR.

[0402] The wireless communication technology in which various embodiments are implemented may include LTE, NR, and 6G, as well as narrowband Internet of things (NB-IoT) for low power communication. For example, the NB-IoT technology may be an example of low power wide area network (LPWAN) technology and implemented as the standards of LTE category (CAT) NB1 and/or LTE Cat NB2. However, these specific appellations should not be construed as limiting NB-IoT. Additionally or alternatively, the wireless communication technology implemented in a wireless device according to various embodiments may enable communication based on LTE-M. For example, LTE-M may be an example of the LPWAN technology, called various names such as enhanced machine type communication (eMTC). For example, the LTE-M technology may be implemented as, but not limited to, at least one of 1) LTE CAT 0, 2) LTE Cat M1, 3) LTE Cat M2, 4) LTE non-bandwidth limited (non-BL), 5) LTE-MTC, 6) LTE machine type communication, and/or 7) LTE M. Additionally or alternatively, the wireless communication technology implemented in a wireless device according to various embodiments may include, but not limited to, at least one of ZigBee, Bluetooth, or LPWAN in consideration of low power communication. For example, ZigBee may create personal area networks (PANs) related to small/low-power digital communication in conformance to various standards such as IEEE 802.15.4, and may be referred to as various names.

[0403] Various embodiments of the disclosure may be implemented by various means. For example, various embodiments of the disclosure may be implemented in hardware, firmware, software, or a combination thereof.

[0404] In a hardware configuration, the methods according to exemplary embodiments of the disclosure may be achieved by one or more Application Specific Integrated Circuits (ASICs), Digital Signal Processors (DSPs), Digital Signal Processing Devices (DSPDs), Programmable Logic Devices (PLDs), Field Programmable Gate Arrays (FPGAs), processors, controllers, microcontrollers, microprocessors, etc.

[0405] In a firmware or software configuration, the methods according to the various embodiments of the disclosure may be implemented in the form of a module, a procedure, a function, etc. performing the above-described functions or operations. A software code may be stored in the memory and executed by the processor. The memory is located at the interior or exterior of the processor and may transmit and receive data to and from the processor via various known means.

[0406] Those skilled in the art will appreciate that the various embodiments of the disclosure may be carried out in other specific ways than those set forth herein without departing from the spirit and essential characteristics of the various embodiments of the disclosure. The above embodiments are therefore to be construed in all aspects as illustrative and not restrictive. The scope of the disclosure should be determined by the appended claims and their legal equivalents, not by the above description, and all changes coming within the meaning and equivalency range of the appended claims are intended to be embraced therein. It is obvious to those skilled in the art that claims that are not explicitly cited in each other in the appended claims may be presented in combination as an embodiment of the disclosure or included as a new claim by a subsequent amendment after the application is filed.

INDUSTRIAL APPLICABILITY

[0407] The various embodiments of disclosure are applicable to various wireless access systems. Examples of the various wireless access systems include a 3GPP system or a 3GPP2 system. Besides these wireless access systems, the various embodiments of the disclosure are applicable to all technical fields in which the wireless access systems find their applications. Moreover, the proposed methods are also applicable to an mmWave communication system using an ultra-high frequency band.

1. A method performed by a user equipment (UE) in a wireless communication system comprising:

receiving assistance data including information related to a configuration for a reference signal (RS) for positioning, wherein the information related to the configuration for the RS includes information related to one or more positioning frequency layers (PFLs); and

performing communication for the RS for positioning based on the assistance data,

wherein, based on information indicating that the UE is a reduced capability (RedCap) UE,

i) the information related to the RS configuration further includes information related to a plurality of first bandwidths (BWs) for the RS; and

ii) the RS is communicated in a single first specific bandwidth (BW) among the plurality of first BWs at a single first time instance,

wherein the plurality of first BWs is included in one first PFL among the one or more PFLs in a frequency domain.

2. The method according to claim **1**, wherein:

based on information indicating that the UE is the RedCap UE and BW-based frequency hopping is configured,

the RS is communicated in the first specific BW at the first time instance, and is communicated in one second specific BW different from the first specific BW among the plurality of first BWs at one second time instance based on the BW-based frequency hopping; and

the second time instance is configured at a time later than the first time instance in a time domain.

3. The method according to claim **2**, wherein:

the plurality of first bandwidths (BWs) is related to a plurality of BW identifiers (IDs); and

in the BW-based frequency hopping, the first specific BW and the second specific BW are identified according to first pattern information based on the plurality of BW IDs.

4. The method according to claim **1**, wherein:

based on the information indicating that the UE is the RedCap UE,

the information related to the RS configuration further includes information related to a plurality of second BWs for the RS; and

the plurality of second BWs is included in the second PFL different from the first PFL among the one or more PFLs within the frequency domain, and

based on information indicating that the UE is the RedCap UE and BW-based frequency hopping is configured,

i) the RS is communicated in the first specific BW at the first time instance, and ii) the RS is communicated in one third specific BW among the plurality of first BWs at one third time instance based on the PFL-based frequency hopping; and

the third time instance is configured at a time later than the first time instance in a time domain.

5. The method according to claim **4**, wherein:

the one or more PFLs are associated with one or more PFL IDs; and

in the PFL-based frequency hopping, the first PFL and the second PFL are identified according to second pattern information based on the one or more PFL IDs.

6. The method according to claim **4**, wherein:

a capability report related to capability of the UE is transmitted; and

based on the information indicating that the UE is the RedCap UE, the capability report includes information related to a maximum number of a plurality of BWs for the RS supportable by the RedCap UE, and the number of the plurality of first BWs and the number of the plurality of second BWs are set to be less than or equal to the maximum number.

7. The method according to claim **1**, wherein:

the plurality of first BWs is configured based on at least one of i), ii), iii) and iv);

wherein

i) means that the plurality of first BWs, in the frequency domain, starts from a lowest physical resource block (PRB) of the first PFL and has a same size;

ii) means that the plurality of first BWs, in the frequency domain, starts from a starting point that is identified based on a common PRB offset configured based on the lowest PRB of the first PFL;

iii) means that the plurality of first BWs, in the frequency domain, starts a starting point that is identified based on the common PRB offset configured based on the lowest PRB of the first PFL, and is configured based on information about a plurality of BW sizes for the plurality of first BWs; and

iv) means that the plurality of first BWs, in the frequency domain, is configured not only based on a plurality of PRB offsets for the plurality of first BWs configured based on the lowest PRB of the first PFL, but also based on information about the plurality of BW sizes for the plurality of first BWs, and is also configured to have the same size.

8. The method according to claim **1**, wherein:

the RS includes at least one of a positioning reference signal (PRS) and a sounding reference signal (SRS) for the positioning.

9. A user equipment (UE) operated in a wireless communication system comprising:

a transceiver; and

at least one processor connected to the transceiver,

wherein the at least one processor is configured to:

receive assistance data including information related to a configuration for a reference signal (RS) for positioning, wherein the information related to the configuration for the RS includes information related to one or more positioning frequency layers (PFLs); and

perform communication for the RS for positioning based on the assistance data,

wherein, based on information indicating that the UE is a reduced capability (RedCap) UE,

i) the information related to the RS configuration further includes information related to a plurality of first bandwidths (BWs) for the RS; and

ii) the RS is communicated in a single first specific bandwidth (BW) among the plurality of first BWs at a single first time instance,

wherein the plurality of first BWs is included in one first PFL among the one or more PFLs in a frequency domain.

10. The user equipment (UE) according to claim **9**, wherein:

the plurality of first BWs is configured based on at least one of i), ii), iii) and iv);

wherein

i) means that the plurality of first BWs, in the frequency domain, starts from a lowest physical resource block (PRB) of the first PFL and has a same size;

ii) means that the plurality of first BWs, in the frequency domain, starts from a starting point that is identified based on a common PRB offset configured based on the lowest PRB of the first PFL;

iii) means that the plurality of first BWs, in the frequency domain, starts a starting point that is identified based on the common PRB offset configured based on the lowest PRB of the first PFL, and is configured based on information about a plurality of BW sizes for the plurality of first BWs; and

iv) means that the plurality of first BWs, in the frequency domain, is configured not only based on a plurality of

PRB offsets for the plurality of first BWs configured based on the lowest PRB of the first PFL, but also based on information about the plurality of BW sizes for the plurality of first BWs, and is also configured to have the same size.

11. The user equipment (UE) according to claim **9**, wherein:

the at least one processor is configured to communicate with at least one of a mobile terminal, a network, and an autonomous vehicle other than a vehicle including the UE.

12. A method performed by a base station (BS) in a wireless communication system comprising:

transmitting assistance data including information related to a configuration for a reference signal (RS) for positioning, wherein the information related to the configuration for the RS includes information related to one or more positioning frequency layers (PFLs); and

performing communication for the RS for positioning related to the assistance data with a user equipment (UE)

wherein, based on information indicating that the UE is a reduced capability (RedCap) UE,

i) the information related to the RS configuration further includes information related to a plurality of first bandwidths (BWs) for the RS; and

ii) the RS is communicated in a single first specific bandwidth (BW) among the plurality of first BWs at a single first time instance,

wherein the plurality of first BWs is included in one first PFL among the one or more PFLs in a frequency domain.

13-15. (canceled)

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